Multiple Linear Regression Modelling of Royal Caribbean Cruise Line's Daily Stock Price

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STP 429

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April 28, 2024

Thank you for the semester, I look forward to DAT 431 :)

Abstract

The goal of this paper is to lay out the analysis of statistical data collected on the end of day prices of RCL (Royal Caribbean Line) stock based on the end of day prices of other stocks in the same market. The best linear regression model can be used to predict RCL's end of day stock. The initial independent variables (company stocks) being considered are: BBY, BWA, CCL, EBAY, EXPE, GRMN, HAS, KMX, LOW, NCLH, NKE, PHM, TSLA, and ULTA.

The first step in the analysis is an exploration of the chosen independent variables. These variables have been chosen based on a series of SAS variable selection procedures. Not every variable has a normal distribution although the total number of outliers appears low. These variables were then individually correlated with the independent variable to confirm that they contributed to the model.

After identifying a potentially significant model we ran a correlation matrix between all numeric variables to determine whether an interaction term would be beneficial. After finding that CCL and EXPE are highly related we made an interaction plot to confirm the possible relevancy of an interaction term. Then we looked at the residual plots accompanying the model; although the plot itself looks relatively scattered, it is obvious that the residual fit plot is not following a linear pattern. To attempt to combat this we added a second order variable based on BBY. We then added these variables to the model and conducted a nested F-test to determine

whether either of them provided prediction power to our model. The F-test came back insignificant so neither of these variables were added to the final model.

After testing a potential interaction term and a potential second order term we wanted to conduct tests for multicollinearity and conduct more in-depth examination of the model's health. By finding the variance inflation or VIF for each variable we were able to remove those causing multicollinearity that could be masking the other variables in the model (PHM, GRMN, CCL). Then we looked at the partial residual plots for the variables in our pared down model. Some variables hug the line better than other variables but nothing looks extremely amiss. Variable transformations could possibly help with this.

Once the model looked ok I ran a final proc reg and removed the only variable with an insignificant p-value (t-test). This did not affect the model much but it did increase the F-value by about 100 so I did not add HAS back in. After the final variables were confirmed I ran more tests on the model. The studentized residual plot revealed that only 3% of the observations in our model rest above or below two standard deviations from the mean. This confirmed the acceptability of the model.

Finally, a linear regression equation was generated and interpreted based on the model: RCL (hat) = 72.94115 + 0.31356(BBY) - 1.57924(BWA) - 0.57643(EBAY) + 0.44319(EXPE) + 0.24021(KMX) + 0.16510(LOW) + 2.64471(NCLH) - 0.20936(NKE) - 0.06695(TSLA) - 0.06111(ULTA) . All regression analysis implies that this is an adequate model. Future testing regarding which corporate sector has the most influence on stocks representing Hotels, Resorts, and Cruise Lines would be interesting. The results from this testing could provide a more generalized way to find relationships between end of day stock prices. If one sector more heavily and consistently influences another sector this could be helpful to know.

Data

The data consists of daily stock prices from 4/1/23 - 3/31/34 for all Consumer Discretionary components of the S&P 500. We were initially observing all 51 components. From this list we wanted to identify which components had the most correlation with our chosen independent component, RCL (Royal Caribbean Cruise Line). To do this we ran a criterion panel and individual selection processes including all components. These iteratively checked the adequacy of models composed of different/different numbers of components and returned, on average, fourteen potentially significant ones that together formed a significant model. Once these variables were identified we ran individual tests to identify their distributions and outliers before paring them down to improve the model. It should also be noted that because they were returned by a variable selection function they all have a p-value that is less than 0.05.

The first chosen component is RCL or Royal Caribbean Cruise Lines, this is our dependent variable and it lives in Hotels, Resorts, and Cruise Lines. The histogram and probability plot show that the distribution of RCL is relatively normal within a certain scope. Beyond this scope it appears as if the distribution has some sort of curvilinear trend. This means that when predicting based off of this data we need to be very sure to only predict within our bounds. The box and whisker plot for RCL implies that there do not appear to be any outliers (fig.17).

The first chosen independent component considered is BBY or Best Buy, this stock lives in Computer and Electronics Retail . The histogram and probability plot show that the distribution of BBY is relatively normal within approximately the same scope as RCL (+/- 2s from mean). Beyond this scope it appears as if the distribution has some sort of curvilinear trend. This means that when predicting based off of this data we need to be very sure to only predict within our bounds. The box and whisker plot for BBY implies that there do not appear to be any outliers. The scatter plot generated between RCL and BBY implies some sort of weak positive linear correlation; this is confirmed by the sign of the parameter estimate and the value of R between these two points (fig.18)(fig.19)(fig.BBYR).

The second chosen independent component considered is BWA or BorgWarner, this stock lives in Automotive Parts and Equipment. The histogram and probability plot show that the distribution of BWA is kind of normal but also not really at all. This data shows an obvious curvilinear trend. This means that its predictive power is limited within a linear regression format but it can still add to the model. The box and whisker plot for BWA implies that there do not appear to be any outliers. The scatter plot generated between RCL and BWA implies the existence of some moderately strong negative linear correlation; this is confirmed by the sign of the parameter estimate and the value of R between these two points (fig.20)(fig.21)(fig.BWAR).

The third chosen independent component considered is CCL or Carnival Cruise Line, this stock lives in Hotels, Resorts, and Cruise Lines. The histogram and probability plot show that the distribution of CCL is kind of normal, better than BWA, but also not really at all. This data shows an obvious curvilinear trend. This means that its predictive power is limited within a linear regression format but it can still add to the model. The box and whisker plot for CCL implies that there do not appear to be any outliers. The scatter plot generated between RCL and CCL implies a strong positive linear correlation; this is confirmed by the sign of the parameter estimate and the value of R between these two points (fig.22)(fig.23)(fig.CCLR).

The fourth chosen independent component considered is EBAY, this stock lives in Broadline Retail. The histogram and probability plot show that the distribution of EBAY is kind of normal but mostly not really at all. This data shows an obvious curvilinear trend. This means that its predictive power is limited within a linear regression format but it can still add to the model. The box and whisker plot for EBAY implies that there are an array of upper outliers and a few lower outliers.do not appear to be any outliers, this isn't wonderful. The scatter plot generated between RCL and EBAY implies some sort of weak positive linear correlation; this is confirmed by the sign of the parameter estimate and the value of R between these two points (fig.24)(fig.25)(fig.EBAYR).

The fifth chosen independent component considered is EXPE, or Expedia, this stock lives in Hotels, Resorts, and Cruise Lines. The histogram and probability plot show that the distribution of EXPE is kind of normal in an even smaller scope than BBY. This data shows an obvious curvilinear trend. This means that its predictive power is limited within a linear regression format but it can still add to the model. The box and whisker plot for EXPE implies that there do not appear to be any outliers. The scatter plot generated between RCL and EXPE implies that there exists a strong positive linear correlation; this is confirmed by the sign of the parameter estimate and the value of R between these two points (fig.26)(fig.27)(fig.EXPER).

The sixth chosen independent component considered is GRMN or Garmin, this stock lives in Consumer Electronics. The histogram and probability plot show that the distribution of GRMN is not really normal at all. This data shows an obvious curvilinear trend. This means that its predictive power is limited within a linear regression format but it can still add to the model. The box and whisker plot for GRMN implies that there do not appear to be any outliers. The scatter plot generated between RCL and GRMN implies that there exists a relatively strong positive linear correlation; this is confirmed by the sign of the parameter estimate and the value of R between these two points (fig.28)(fig.29)(fig.GRMNR)..

The seventh chosen independent component considered is HAS, this stock lives in Leisure Products. The histogram and probability plot show that the distribution of HAS is actually pretty good looking all things considered.. This data shows an obvious curvilinear trend but hugs the normal quantile lines better than most of the other variables. This means that its predictive power is limited within a linear regression format but it can still add to the model, perhaps even more so than its less normal counterparts. The box and whisker plot for HAS implies that there do not appear to be any outliers. The scatter plot generated between RCL and HAS implies some sort of weak negative linear correlation; this is confirmed by the sign of the parameter estimate and the value of R between these two points (fig.30)(fig.31)(fig.HASR).

The eighth chosen independent component considered is KMX or CarMax, this stock lives in Automotive Retail. The histogram and probability plot show that the distribution of KMX is kind of normal, it looks very similar to the distribution of HAS. This means that its predictive power is limited within a linear regression format but it can still add to the model, perhaps even more so than its less normal counterparts. The box and whisker plot for KMX implies that there do not appear to be any outliers.. The scatter plot generated between RCL and KMX implies some sort of weak positive linear correlation; this is confirmed by the sign of the parameter estimate and the value of R between these two points (fig.32)(fig.33)(fig.KMXR).

The ninth chosen independent component considered is LOW or Lowe's, this stock lives in Home Improvement Retail. The histogram and probability plot show that the distribution of LOW is pretty normal within 2s of the mean. The histogram and probability plot show that the distribution of LOW is relatively normal within approximately the same scope as RCL (+/- 2s from mean). Beyond this scope it appears as if the distribution has some sort of curvilinear trend. This means that when predicting based off of this data we need to be very sure to only predict within our bounds. The box and whisker plot for LOW implies that there exists a single upper outlier. The scatter plot generated between RCL and LOW implies that there exists some moderately strong positive linear correlation; this is confirmed by the sign of the parameter estimate and the value of R between these two points (fig.34)(fig.35)(fig.LOWR).

The tenth chosen independent component considered is NCLH or Norwegian Cruise Line Holding, this stock lives in Hotels, Resorts, and Cruise Lines. The histogram and probability plot show that the distribution of NCLH is kind of normal but not really. It does look better than some of the other components but there is a distinct curvilinear trend. The box and whisker plot for NCLH implies that there do not appear to be any outliers. The scatter plot generated between RCL and NCLH implies that there exists some moderately strong positive linear correlation; this is confirmed by the sign of the parameter estimate and the value of R between these two points (fig.36)(fig.37)(fig.NCLHR).

The eleventh chosen independent component considered is NKE or Nike, this stock lives in Apparel, Accessories, and Commercial Goods. The histogram and probability plot show that the distribution of NKE is kind of normal but has a distinct curvilinear trend This means that NKE could still contribute to the model but it might not be best represented by a linear regression model. The box and whisker plot for NKE implies that there exists an array of upper outliers. The scatter plot generated between RCL and NKE implies some sort of moderately weak negative linear correlation; this is confirmed by the sign of the parameter estimate and the value of R between these two points (fig.38)(fig.39)(fig.RNKE).

The twelfth chosen independent component considered is PHM or PulteGroup, this stock lives in Home Building. The histogram and probability plot show that the distribution of PHM is not consistently normal and has a distinct curvilinear trend. The box and whisker plot for PHM implies that there do not appear to be any outliers. The scatter plot generated between RCL and PHM implies that there exists a very strong positive correlation, this is confirmed by the sign of the parameter estimate and the value of R^2 between these two points (fig.40)(fig.41)(fig.PHMR).

The thirteenth chosen independent component considered is TSLA, this stock lives in Automobile Manufacturers. The histogram and probability plot show that the distribution of TSLA is relatively normal within approximately the same scope as RCL (+/- 2s from mean). The data does appear primarily curvilinear though, this means that while it might contribute to our model it may be best represented by a non-linear model. The box and whisker plot for TSLA implies that there do not appear to be any outliers. The scatter plot generated between RCL and TSLA implies some sort of really weak positive linear correlation; this is confirmed by the sign of the parameter estimate and the value of R^2 between these two points (fig.42)(fig.43)(fig.TSLAR).

The final chosen independent component considered is ULTA, this stock lives in Other Speciality Retail. The histogram and probability plot show that the distribution of ULTA is relatively normal within approximately the same scope as RCL (+/- 2s from mean) but also not really, as there appears a distinct curvilinear trend. The box and whisker plot for ULTA implies that there do not appear to be any outliers. The scatter plot generated between RCL and ULTA implies some sort of weak positive linear correlation; this is confirmed by the sign of the parameter estimate and the value of R^2 between these two points (fig.44)(fig.45)(ULTAR).

After finding the best first order model we wanted to look for potential interaction terms and second order terms. These were identified and tested using various SAS procedures. To perform this analysis we had to generate a new data set containing the potential interaction and second order terms. Their significance was then determined and they were not added to our model. Once this was confirmed we worked to reduce any multicollinearity in the model using various SAS procedures. This led to us removing three additional variables from our model. Then we re-ran a linear regression model on the data and removed one component that was no longer returning a significant p-value to reach our final model.

Methodology

Before beginning our analysis we had to find how many observations we were working with and we had to identify the variable's taxonomy. To do this we ran proc contents in SAS (fig. 01)(fig.02). After we established what our dataset looked like we ran a proc reg on all 51 components including the stepwise selection function (fig.03). This function ran through all potential models and eliminated or added variables based on how those terms impacted the R^2 and C(p) of the model. The stepwise regression function returned 18 variables with each of them significant at the 0.05 level (fig.04). We wanted to see if other variable selection methods would return a similar "best model".

The next variable selection method we used was R^2. This method returned the five best models containing 8-20 variables (fig.05). The R^2 does not appear to increase dramatically at any point after approximately 9 variables are included in the model (fig.06). This implies that the best model probably contains around 9 variables. We then wanted to use the R^2 adj variable selection method (fig.07). The highest R^2 adj value we can achieve is 0.9948 with approx. 35 variables (fig.08). We may not need to aim so high but at least we know we can achieve a good model. We also used the variable selection methods that are based on C(p) (fig.09). This method focuses on minimizing the mean square error of the model and regression bias. The smallest achievable C(p) is approx. 21, this occurs with 29 variables (fig.10). The final selection method we used before looking at them together utilizes the press criterion to determine a good model. This method uses the predictive sum of squares to identify which model has the most predictive power. This method cannot be run in a proc reg and must be within a proc glmselect (fig.11). The optimal press criterion is the smallest one possible. This was identified with a fourteen variable model (fig.12).

We then used the GLMSELECT procedure to compare the variable selection methods (fig.13). The criterion panel implied that, in general, each selection method agreed on a 20 variable model (fig.14). The final selection used many more stats to produce a fourteen variable model (fig.15). We proceeded using this model for the rest of the analysis.

After identifying our variables we ran proc univariate functions which graphed a histogram and box-and-whiskers plots from which we were able to discern distribution and potential outliers; we simultaneously performed a proc sgplot function to get an idea of what the correlation between our independent variables and wins looks like (fig.16)(fig.17).

Once we had identified and were familiar with our dataset and data we wanted to test if an interaction term or second order term would benefit the model. Starting with the interaction term, we ran a proc corr between all numeric variables in the dataset. CCL and EXPE appeared highly, and significantly correlated (fig.46)(fig.47). To confirm this we created an interaction plot using proc glm/plm. The interaction plot had non-parallel lines so we added a CCL_EXPE variable into a second data set to be tested for significance (fig.48)(fig.49). We then ran a proc reg function to look at the residual plot for our variables, it did not appear random so we decided to create and test a second order term. BBY_SQUARED was added to the new data set (fig.50)(fig.51). Then a nested F-test was performed including both new variables. The nested

F-test did not return significantly which implies the inclusion of these variables is not necessary and they were not added to the model (fig.52)(fig.53).

After finding the best model we ran a proc reg test including VIF. The stocks with VIF > 10 were then removed one at a time until there was no potential for one variable to mask another (fig.54)(fig.55). Then we ran residual plots using proc reg to determine whether there appeared to be any heteroscedasticity or unequal variances in the model. You want each variable to vary equally from the LSRL so this is not ideal if it shows up. The residual plots were not completely linear, which would imply equal variances but they looked pretty good so I decided to leave it alone (fig.56)(fig.57). Finally we ran a proc reg test that included and printed all additional stats. We did this to confirm that the studentized residual plot did not show an unreasonable percentage of outliers (fig.58)(fig.59). After checking that everything in the model looks good and removing a final variable that did not have a significant p-value, we used our data to find the least squares regression line equation (fig.60).

Results

Model	F-value	CoeffVar	R^2 (adj)	Overall Adequacy
RCL(hat) = B0 + B1(BBY) + B2(BWA) + B3(CCL) + B4(EBAY) + B5(EXPE) + B6(GRMN) + B7(HAS) + B8(KMX) + B9(LOW) + B10(NCLH) + B11(NKE) + B12(PHM) + B13(TSLA) + B14(ULTA) + error	2506.85 (<0.0001)	1.63617	0.9930	Great but: Interaction terms present, residual plot is not random, multicollinearity is unchecked
RCL(hat) = B0 + B1(BBY) + B2(BWA) + B3(CCL) + B4(EBAY) + B5(EXPE) + B6(GRMN) + B7(HAS) + B8(KMX) + B9(LOW) + B10(NCLH) + B11(NKE) +	2183.56 (<0.0001)	1.64045	0.9929	Still Great but: New terms probably not necessary (no big change in model), multicollinearity is unchecked

B12(PHM) + B13(TSLA) + B14(ULTA) + B15(CCL)(EXPE) + B16(BBY^2) + error				
RCL(hat) = B0 + B1(BBY) + B2(BWA) + B4(EBAY) + B5(EXPE) + B7(HAS) + B8(KMX) + B9(LOW) + B10(NCLH) + B11(NKE) + B13(TSLA) + B14(ULTA) + error	1181.70 (<0.0001)	2.67405	0.9813	Looking Good: no more multicollinearity but one of my p-values is insignificant
RCL(hat) = B0 + B1(BBY) + B2(BWA) + B4(EBAY) + B5(EXPE) + B8(KMX) + B9(LOW) + B10(NCLH) + B11(NKE) + B13(TSLA) + B14(ULTA) + error	1291.56 (<0.0001)	2.68238	0.9811	Still looks really good: all p-values significant and multicollinearity checked

The first row details information pertaining to the fourteen variable model returned by our various selection processes. This model returned an F-value of 2506.85 which is highly significant. This means that we have enough evidence to reject the null hypothesis that B1-14=0 and instead support the conclusion that at least one of the variables in this model contributes to our ability to predict the price of RCL. The coefficient of variation (CV) in this model is 1.63. This is really good, it means that the data in this model varies very slightly from the LSRL, implying strength in the predictive power of our LSRL. Any CV value less than or equal to approx. 10 contributes to the implication of a strong LSRL. Finally the R^2 adj is 0.9930. This means that approximately 99.3% of the variation in RCL can be accounted for by the variables in our model. This is basically as mouth-watering as you can get in terms of an R^2 adj value. The model stats look amazing but we still need to test whether the addition of an interaction and second order term is necessary.

These were added into the model tested in row 2. The stats look good but have not significantly improved. This implies that the additional variables are probably not necessary. This is confirmed by a nested f-test which returned an insignificant f-value of 0.47. This means that we do not have enough evidence to reject the null hypothesis that the addition of an interaction or second order variable was beneficial to the model (B15=B16=0). After confirming that these terms were unnecessary we removed them and tested the original fourteen-variable model for multicollinearity.

Three variables, PHM, GRMN, and CCL returned VIF values of approx. 46, 28, and 27 respectively. A variance inflation value that is higher than 10 implies that the corresponding variable is correlated so highly with another that their relationship is hiding the true nature of the relationship between our chosen dependent and independent variables. These variables were subsequently removed and the proc reg function was performed resulting in the data from row 3. The key statistics still look good and the model is now multicollinearity-proof. However, upon closer inspection there is a variable, HAS, that does not return a significant p-value. This means that we have enough evidence to support the null hypothesis that B5=0 and reject the idea that B5, which determines how much HAS impacts RCL, has any value. This means that there is not enough evidence to conclude that this variable has any effect on RCL and it can be removed from the model.

The final model represented in row 4 reflects the final 10 variables. The model returns a significant F-value of 1291.56, it should also be noticed that the F-value improved by approx. 100 points simply from the removal of HAS. A significant F-value speaks highly to the adequacy of the model. The model returns a CV of 2.68 which is still much lower than 10 (good). It also has an R^2 (adj) value of 0.9811. This means that 98% of the variance in RCL can be explained by our variables; this is very, very good. It appears that we have our best model but we want to further scrutinize where we landed. To do this we looked at the partial residual plots for each variable in the model. In an ideal model the data for each variable would follow a linear pattern. Although some of the plots look more linear than others none of them look entirely random or blatantly non-linear. I tried various transformations on both the independent and dependent variables to try to straighten these plots out but there were no transformations that significantly improved the data. After concluding that the partial residuals looked acceptable I ran a test to look at the studentized residual plot and the cooksd. Although the cooksd contains some outliers the studentized residual plot has highlighted any problematic residuals associated with them. Only 3% of observations are problematic; this is acceptable. All relevant residual and fit plots have been analyzed and they do not significantly question the validity of our model. This implies that we have reached the best possible model with row 4.

Our best model: RCL (hat) = 72.94115 + 0.31356(BBY) - 1.57924(BWA) - 0.57643(EBAY) + 0.44319(EXPE) + 0.24021(KMX) + 0.16510(LOW) + 2.64471(NCLH) - 0.20936(NKE) - 0.06695(TSLA) - 0.06111(ULTA) implies that if none of our independent variables were factors then we would expect the daily RCL stock price to be approx. $72.94. We know that for every one unit increase in BBY stock price we expect an $0.31 increase in RCL. For every one unit increase in BWA we expect an $1.57 decrease in RCL. For every one unit increase in EBAY we expect an $0.57 decrease in RCL. For every one unit increase in EXPE we expect an $0.44 increase in RCL. For every one unit increase in KMX we expect an $0.24 increase in RCL. For every one unit increase in LOW we expect an $0.16 increase in RCL. For every one unit increase in NCLH we expect an $2.65 increase in RCL. For every one unit increase in NKE we expect an $0.21 increase in RCL. For every one unit increase in TSLA we expect a $0.07 decrease in RCL. For every one unit increase in ULTA we expect a $0.06 decrease in RCL.

Conclusion

We can conclude that between 4/1/23 - 3/31/34 the daily stock price for Royal Caribbean Cruise Line can best be predicted by the daily prices of BestBuy, BorgWarner, Ebay, Experian, CarMax, Lowe's, Norwegian Cruise Line Holdings, Tesla, and Ulta Beauty. This model is highly significant, has acceptable relevant statistics, fit plots, and residual plots. All of this suggests that we can very accurately predict the stock price of RCL using our model. In terms of next steps I would like to continue to experiment with variable transformations. I think that the data could be relevant within a larger scope if I could figure out where to tweak. I would start by creating log, inverse, and arcsin versions of every variable, then I would go through the ideal of testing the significance of each hypothetical addition. This might make the partial residual fit plots look more linear, it also might make the residual plot look more scattered. At the same time, significant transformations might be highly detrimental to the integrity of the data. I would also want to continue to explore my original question about whether component types have relationships as opposed to simply component themselves having relationships. Out of the ten final variables used in the model 20% came from Hotels, Resorts, and Cruise Lines; 30% came from the automobile industry, and 60% were in retail. Further analysis into the relationship between stock price, stock, and industry would be pretty cool but I would want it to be in 3D and that's a little outside of MY scope. These results could determine whether certain industries have daily stock price relationships which reveals more about potential relationships between the industries themselves.

Appendix

01

```
FILENAME REFFILE '/home/u63735632/sasuser.v94/Spring 2024 Lab #3 Closing Prices (1).xlsx';

PROC IMPORT DATAFILE=REFFILE
    DBMS=XLSX
    OUT=WORK.prices;
    GETNAMES=YES;
RUN;

PROC CONTENTS DATA=WORK.prices; RUN;


%web_open_table(WORK.IMPORT);
```

02

The CONTENTS Procedure

Data Set Name	WORK.PRICES	**Observations**	249
Member Type	DATA	**Variables**	53
Engine	V9	**Indexes**	0
Created	04/27/2024 20:50:32	**Observation Length**	424
Last Modified	04/27/2024 20:50:32	**Deleted Observations**	0
Protection		**Compressed**	NO
Data Set Type		**Sorted**	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Alphabetic List of Variables and Attributes

#	Variable	Type	Len	Format	Label
2	ABNB	Num	8	BEST.	ABNB
3	AMZN	Num	8	BEST.	AMZN
4	APTV	Num	8	BEST.	APTV
5	AZO	Num	8	BEST.	AZO
6	BBWI	Num	8	BEST.	BBWI
7	BBY	Num	8	BEST.	BBY
8	BKNG	Num	8	BEST.	BKNG
9	BWA	Num	8	BEST.	BWA
12	CCL	Num	8	BEST.	CCL
13	CMG	Num	8	BEST.	CMG
10	CZR	Num	8	BEST.	CZR
15	DECK	Num	8	BEST.	DECK
17	DHI	Num	8	BEST.	DHI
16	DPZ	Num	8	BEST.	DPZ
14	DRI	Num	8	BEST.	DRI
1	Date	Num	8	MMDDYY10.	Date
18	EBAY	Num	8	BEST.	EBAY
19	ETSY	Num	8	BEST.	ETSY
20	EXPE	Num	8	BEST.	EXPE
21	F	Num	8	BEST.	F
23	GM	Num	8	BEST.	GM
24	GPC	Num	8	BEST.	GPC
22	GRMN	Num	8	BEST.	GRMN
25	HAS	Num	8	BEST.	HAS
27	HD	Num	8	BEST.	HD
26	HLT	Num	8	BEST.	HLT
11	KMX	Num	8	BEST.	KMX
29	LEN	Num	8	BEST.	LEN
30	LKQ	Num	8	BEST.	LKQ
31	LOW	Num	8	BEST.	LOW
32	LULU	Num	8	BEST.	LULU
28	LVS	Num	8	BEST.	LVS
33	MAR	Num	8	BEST.	MAR
34	MCD	Num	8	BEST.	MCD
35	MGM	Num	8	BEST.	MGM
36	MHK	Num	8	BEST.	MHK
38	NCLH	Num	8	BEST.	NCLH
37	NKE	Num	8	BEST.	NKE
39	NVR	Num	8	BEST.	NVR
40	ORLY	Num	8	BEST.	ORLY
42	PHM	Num	8	BEST.	PHM
41	POOL	Num	8	BEST.	POOL
45	RCL	Num	8	BEST.	RCL
43	RL	Num	8	BEST.	RL
44	ROST	Num	8	BEST.	ROST
46	SBUX	Num	8	BEST.	SBUX
49	TJX	Num	8	BEST.	TJX
47	TPR	Num	8	BEST.	TPR
50	TSCO	Num	8	BEST.	TSCO
48	TSLA	Num	8	BEST.	TSLA
51	ULTA	Num	8	BEST.	ULTA
52	WYNN	Num	8	BEST.	WYNN
53	YUM	Num	8	BEST.	YUM

03

```
**runs stepwise regression**;
proc reg data=prices;
model RCL = ABNB AMZN APTV AZO BBWI BBY BKNG BWA CCL CMG CZR DECK DHI DPZ DRI EBAY ETSY EXPE F
GM GPC GRMN HAS HD HLT KMX LEN LKQ LOW LULU LVS MAR MCD MGM MHK NCLH NKE NVR ORLY PHM POOL RL ROST
SBUX TJX TPR TSCO TSLA ULTA WYNN YUM/selection=stepwise slstay=0.05;
run;
```

04

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	8.34907	8.00943	2.73443	1.09	0.2983
BBY	0.32404	0.07004	53.86516	21.40	<.0001
BKNG	0.00443	0.00133	28.15339	11.19	0.0010
BWA	-0.53052	0.08500	98.02493	38.95	<.0001
CCL	1.93097	0.20944	213.90083	85.00	<.0001
EBAY	-0.37540	0.12744	21.83667	8.68	0.0036
EXPE	0.12138	0.02310	69.45714	27.60	<.0001
GRMN	0.30850	0.04399	123.73960	49.17	<.0001
HAS	0.22374	0.04333	67.09828	26.66	<.0001
KMX	0.10732	0.04991	11.63448	4.62	0.0326
LOW	-0.07041	0.02813	15.76236	6.26	0.0130
MCD	0.07582	0.02348	26.24182	10.43	0.0014
MHK	-0.18921	0.05058	35.21129	13.99	0.0002
NCLH	1.17798	0.17123	119.10111	47.33	<.0001
NKE	-0.14970	0.02405	97.53739	38.76	<.0001
PHM	0.38983	0.05627	120.78964	48.00	<.0001
TPR	0.14312	0.05872	14.94765	5.94	0.0156
TSLA	-0.07199	0.00840	184.85205	73.46	<.0001
ULTA	-0.04464	0.00653	117.58818	46.73	<.0001

Bounds on condition number: 78.969, 6739

All variables left in the model are significant at the 0.0500 level.

The stepwise method terminated because the next variable to be entered was just removed.

05

```
**runs r^2**;
proc reg data=prices;
model RCL = ABNB AMZN APTV AZO BBWI BBY BKNG BWA CCL CMG CZR DECK DHI DPZ DRI EBAY ETSY EXPE F
GM GPC GRMN HAS HD HLT KMX LEN LKQ LOW LULU LVS MAR MCD MGM MHK NCLH NKE NVR ORLY PHM POOL RL ROST
SBUX TJX TPR TSCO TSLA ULTA WYNN YUM/selection=rsquare start=8 stop=20 best=5;
run;
```

06

6	0.9887	APTV CCL DPZ MAR NKE PHM
6	0.9887	APTV CCL GM HAS MHK PHM
6	0.9887	BWA CCL DPZ GRMN NKE PHM
7	0.9898	APTV CCL EXPE HAS NCLH NKE PHM
7	0.9897	APTV CCL EXPE GRMN HAS PHM SBUX
7	0.9896	APTV CCL DPZ GRMN MAR NKE PHM
7	0.9896	APTV CCL DRI GM HAS MHK PHM
7	0.9895	APTV CCL DHI EXPE KMX PHM SBUX
8	0.9905	APTV CCL EXPE HAS LEN NCLH NKE PHM
8	0.9904	BWA CCL EXPE HAS NCLH PHM TSLA ULTA
8	0.9904	BWA CCL EXPE NCLH NKE PHM TSLA ULTA
8	0.9903	APTV CCL DPZ EXPE GRMN HAS NKE PHM
8	0.9903	APTV CCL EXPE KMX LEN NCLH PHM SBUX
9	0.9913	BWA CCL EXPE HAS NCLH NKE PHM TSLA ULTA
9	0.9912	BWA CCL EXPE GRMN HAS NCLH PHM TSLA ULTA
9	0.9911	BWA CCL EXPE KMX NCLH NKE PHM TSLA ULTA
9	0.9910	BWA CCL EXPE GRMN KMX PHM SBUX TSLA ULTA
9	0.9910	APTV CCL EXPE GRMN HAS PHM SBUX TSLA ULTA
10	0.9920	BWA CCL EXPE GRMN HAS NCLH NKE PHM TSLA ULTA
10	0.9919	BWA CCL EXPE GRMN HAS PHM SBUX TSLA ULTA YUM
10	0.9918	BWA CCL EXPE GRMN HAS NCLH PHM SBUX TSLA ULTA
10	0.9918	BWA CCL EXPE GRMN HAS PHM SBUX TPR TSLA ULTA
10	0.9917	APTV CCL EXPE GM GRMN HAS MHK PHM TSLA ULTA
11	0.9924	BWA CCL EBAY EXPE GRMN HAS NCLH NKE PHM TSLA ULTA
11	0.9924	AZO BWA CCL EXPE GRMN HAS NCLH NKE PHM TSLA ULTA
11	0.9923	BWA CCL EXPE GRMN HAS MCD NCLH PHM SBUX TSLA ULTA
11	0.9923	BBWI BWA CCL EXPE GRMN HAS NCLH NKE PHM TSLA ULTA
11	0.9923	BWA CCL EXPE GRMN HAS NCLH PHM SBUX TSLA ULTA YUM
12	0.9929	BBY BWA CCL EBAY EXPE GRMN HAS NCLH NKE PHM TSLA ULTA

07

```
**adj r^2 criterion**;
proc reg data=prices;
model RCL = ABNB AMZN APTV AZO BBWI BBY BKNG BWA CCL CMG CZR DECK DHI DPZ DRI EBAY ETSY EXPE F GM
GPC GRMN HAS HD HLT KMX LEN LKQ LOW LULU LVS MAR MCD MGM MHK NCLH NKE NVR ORLY PHM POOL RL ROST
SBUX TJX TPR TSCO TSLA ULTA WYNN YUM/selection=adjrsq;
run;
```

08

The REG Procedure
Model: MODEL1
Dependent Variable: RCL

Adjusted R-Square Selection Method

Number of Observations Read	249
Number of Observations Used	249

Number in Model	Adjusted R-Square	R-Square	Variables in Model
34	0.9948	0.9956	ABNB APTV AZO BBY BKNG BWA CCL DECK DRI EBAY EXPE F GM GRMN HD KMX LEN LKQ MAR MCD MGM MHK NCLH NKE ORLY PHM POOL RL ROST TPR TSCO TSLA ULTA WYNN
36	0.9948	0.9956	ABNB AMZN APTV AZO BBY BKNG BWA CCL CMG DECK DRI EBAY EXPE F GM GRMN HD KMX LEN LKQ MAR MCD MGM MHK NCLH NKE ORLY PHM POOL RL ROST TPR TSCO TSLA ULTA WYNN
35	0.9948	0.9956	ABNB AMZN APTV AZO BBY BKNG BWA CCL DECK DRI EBAY EXPE F GM GRMN HD KMX LEN LKQ MAR MCD MGM MHK NCLH NKE ORLY PHM POOL RL ROST TPR TSCO TSLA ULTA WYNN
33	0.9948	0.9955	ABNB APTV AZO BBY BKNG BWA CCL DECK DRI EXPE F GM GRMN HD KMX LEN LKQ MAR MCD MGM MHK NCLH NKE ORLY PHM POOL RL ROST TPR TSCO TSLA ULTA WYNN
35	0.9948	0.9956	ABNB AMZN APTV AZO BBY BKNG BWA CCL CMG DECK DRI EBAY EXPE F GM GRMN HD KMX LEN LKQ MAR MCD MGM MHK NCLH NKE ORLY PHM POOL RL ROST TPR TSCO TSLA ULTA
35	0.9948	0.9956	ABNB APTV AZO BBY BKNG BWA CCL CZR DECK DRI EBAY EXPE F GM GRMN HD KMX LEN LKQ MAR MCD MGM MHK NCLH NKE ORLY PHM POOL RL ROST TPR TSCO TSLA ULTA WYNN
35	0.9948	0.9956	ABNB APTV AZO BBY BKNG BWA CCL CMG DECK DRI EBAY EXPE F GM GRMN HD KMX LEN LKQ MAR MCD MGM MHK NCLH NKE ORLY PHM POOL RL ROST TPR TSCO TSLA ULTA WYNN
37	0.9948	0.9956	ABNB AMZN APTV AZO BBY BKNG BWA CCL CMG CZR DECK DRI EBAY EXPE F GM GRMN HD KMX LEN LKQ MAR MCD MGM MHK NCLH NKE ORLY PHM POOL RL ROST TPR TSCO TSLA ULTA WYNN
34	0.9948	0.9955	ABNB AMZN APTV AZO BBY BKNG BWA CCL DECK DRI EBAY EXPE F GM GRMN HD KMX LEN LKQ MAR MCD MGM MHK NCLH NKE ORLY PHM POOL RL ROST TPR TSCO TSLA ULTA
34	0.9948	0.9955	ABNB AMZN APTV AZO BBY BKNG BWA CCL DECK DRI EXPE F GM GRMN HD KMX LEN LKQ MAR MCD MGM MHK NCLH NKE ORLY PHM POOL RL ROST TPR TSCO TSLA ULTA WYNN
33	0.9948	0.9955	ABNB APTV AZO BBY BKNG BWA CCL DECK DRI EBAY EXPE F GM GRMN HD KMX LEN LKQ MAR MCD MGM MHK NCLH NKE ORLY PHM POOL RL ROST TPR TSCO TSLA ULTA
35	0.9948	0.9956	ABNB APTV AZO BBY BKNG BWA CCL DECK DRI EBAY EXPE F GM GRMN HD KMX LEN LKQ MAR MCD MGM MHK NCLH NKE ORLY PHM POOL RL ROST TPR TSCO TSLA ULTA WYNN YUM

09

```
**cp**;
proc reg data=prices;
model RCL = ABNB AMZN APTV AZO BBWI BBY BKNG BWA CCL CMG CZR DECK DHI DPZ DRI EBAY ETSY EXPE F GM
GPC GRMN HAS HD HLT KMX LEN LKQ LOW LULU LVS MAR MCD MGM MHK NCLH NKE NVR ORLY PHM POOL RL ROST
SBUX TJX TPR TSCO TSLA ULTA WYNN YUM/selection=cp;
run;
```

10

The REG Procedure
Model: MODEL1
Dependent Variable: RCL

C(p) Selection Method

Number of Observations Read	249
Number of Observations Used	249

Number in Model	C(p)	R-Square	Variables in Model
29	21.3636	0.9954	ABNB AZO BBY BKNG BWA CCL DECK DRI EBAY EXPE F GM GRMN HD KMX LEN MAR MCD MGM MHK NCLH NKE ORLY PHM RL TPR TSCO TSLA ULTA

11

```
***press statistic**;
proc glmselect data=prices;
model RCL = ABNB AMZN APTV AZO BBWI BBY BKNG BWA CCL CMG CZR DECK DHI DPZ DRI EBAY ETSY EXPE F GM
GPC GRMN HAS HD HLT KMX LEN LKQ LOW LULU LVS MAR MCD MGM MHK NCLH NKE NVR ORLY PHM POOL RL ROST
SBUX TJX TPR TSCO TSLA ULTA WYNN YUM/selection=stepwise(choose=press);
run;
```

12

The GLMSELECT Procedure

Stepwise Selection Summary

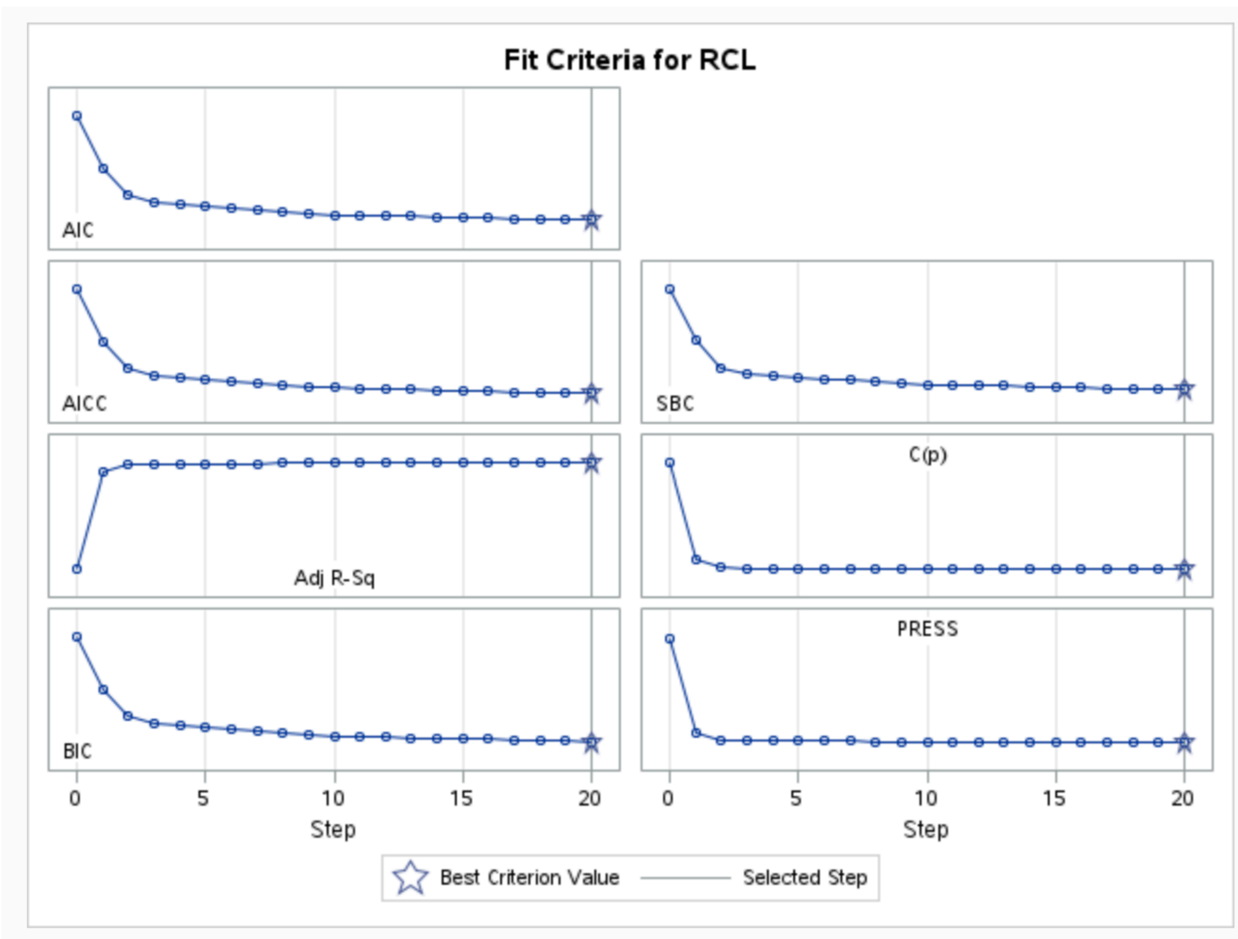
Step	Effect Entered	Effect Removed	Number Effects In	SBC	PRESS
0	Intercept		1	1490.8516	97801.8112
1	PHM		2	891.9268	8711.0415
2	CCL		3	573.9756	2400.6404
3	MGM		4	488.0897	1675.7500
4	CMG		5	473.9888	1567.1514
5	ULTA		6	454.4088	1423.5042
6	BBWI		7	434.7124	1295.1020
7	TSLA		8	421.2634	1210.9186
8	EXPE		9	399.1670	1090.6584
9	HAS		10	386.6235	1022.1783
10	GRMN		11	369.2569	937.3534
11	LOW		12	367.3225	916.7484
12	BBY		13	365.9186	900.9403
13	BWA		14	360.5993	869.3935
14	NKE		15	349.3419	824.9983
15		MGM	14	348.5632	832.7143
16		CMG	13	347.8597	840.8779
17	NCLH		14	331.4169	778.1750
18	EBAY		15	325.1597	751.4817
19		BBWI	14	323.6162	757.9676
20	KMX		15	318.8219*	733.1698*

*** Optimal Value of Criterion**

13

```
proc glmselect data=prices plots=criterionpanel;
model RCL = ABNB AMZN APTV AZO BBWI BBY BKNG BWA CCL CMG CZR DECK DHI DPZ DRI EBAY ETSY EXPE F GM
GPC GRMN HAS HD HLT KMX LEN LKQ LOW LULU LVS MAR MCD MGM MHK NCLH NKE NVR ORLY PHM POOL RL ROST
SBUX TJX TPR TSCO TSLA ULTA WYNN YUM/selection=stepwise stats=all;
run;
```

14



15

The GLMSELECT Procedure
Selected Model

The selected model is the model at the last step (Step 20).

Effects:	Intercept BBY BWA CCL EBAY EXPE GRMN HAS KMX LOW NCLH NKE PHM TSLA ULTA

Analysis of Variance				
Source	DF	Sum of Squares	Mean Square	F Value
Model	14	96375	6883.94683	2506.85
Error	234	642.57630	2.74605	
Corrected Total	248	97018		

Root MSE	1.65712
Dependent Mean	101.24711
R-Square	0.9934
Adj R-Sq	0.9930
AIC	517.06013
AICC	519.40496
BIC	262.82333
C(p)	81.93352
PRESS	733.16983
SBC	318.82193
ASE	2.58063

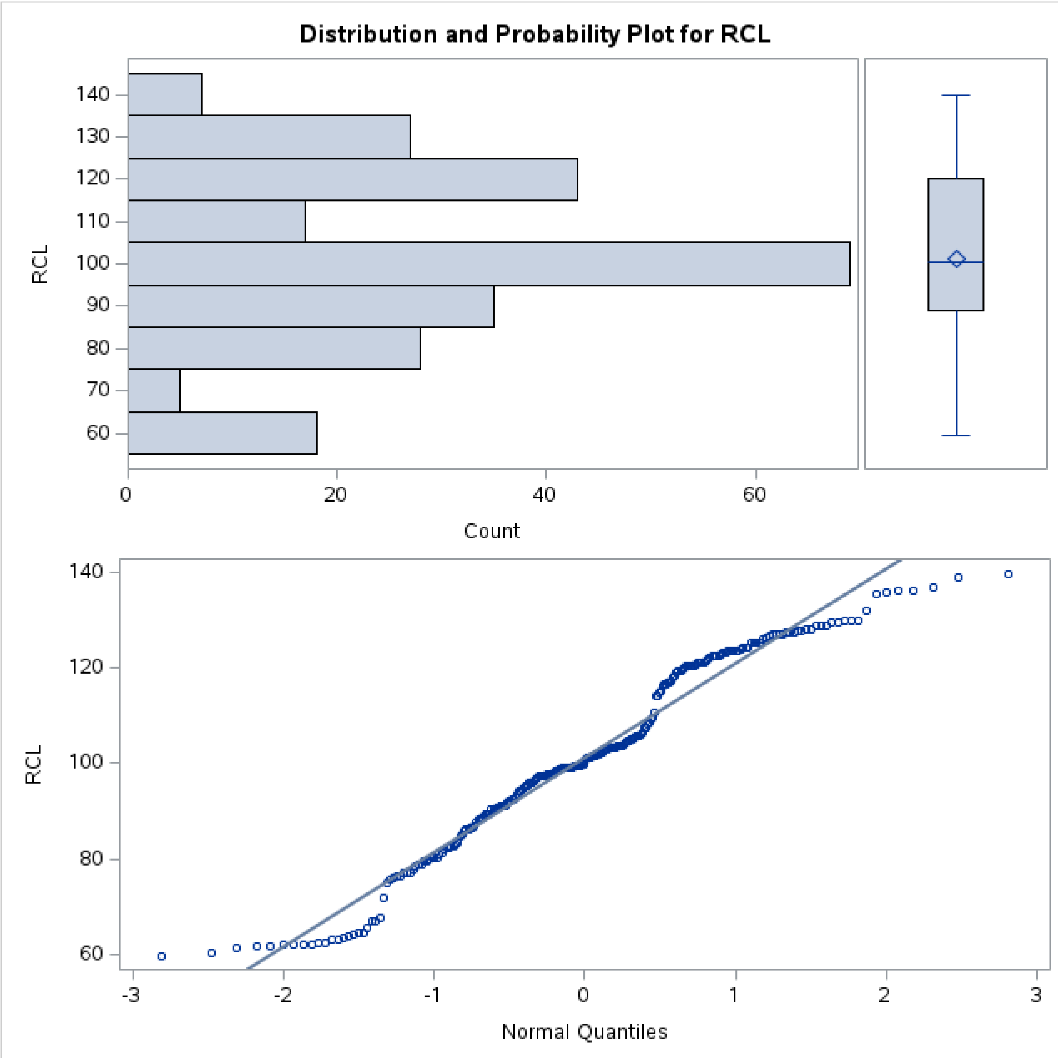
Parameter Estimates				
Parameter	DF	Estimate	Standard Error	t Value
Intercept	1	34.912334	5.516691	6.33
BBY	1	0.253823	0.063367	4.01
BWA	1	-0.590551	0.087185	-6.77
CCL	1	2.082130	0.203183	10.25
EBAY	1	-0.566537	0.125936	-4.50
EXPE	1	0.175717	0.019791	8.88
GRMN	1	0.323561	0.043278	7.48
HAS	1	0.285926	0.041169	6.95
KMX	1	0.152592	0.048512	3.15
LOW	1	-0.099858	0.026911	-3.71
NCLH	1	0.980707	0.155588	6.30
NKE	1	-0.155492	0.022588	-6.88
PHM	1	0.366621	0.044876	8.17
TSLA	1	-0.069804	0.008154	-8.56
ULTA	1	-0.032231	0.005224	-6.17

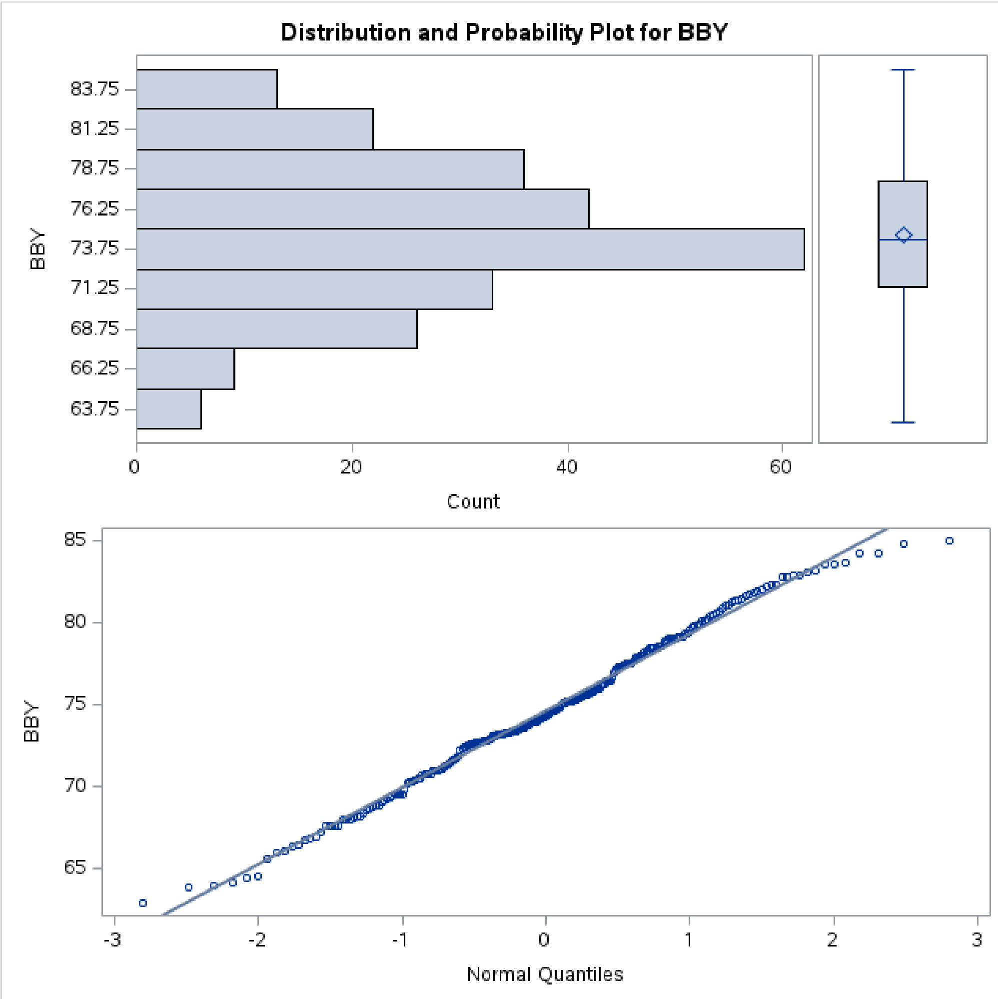
16

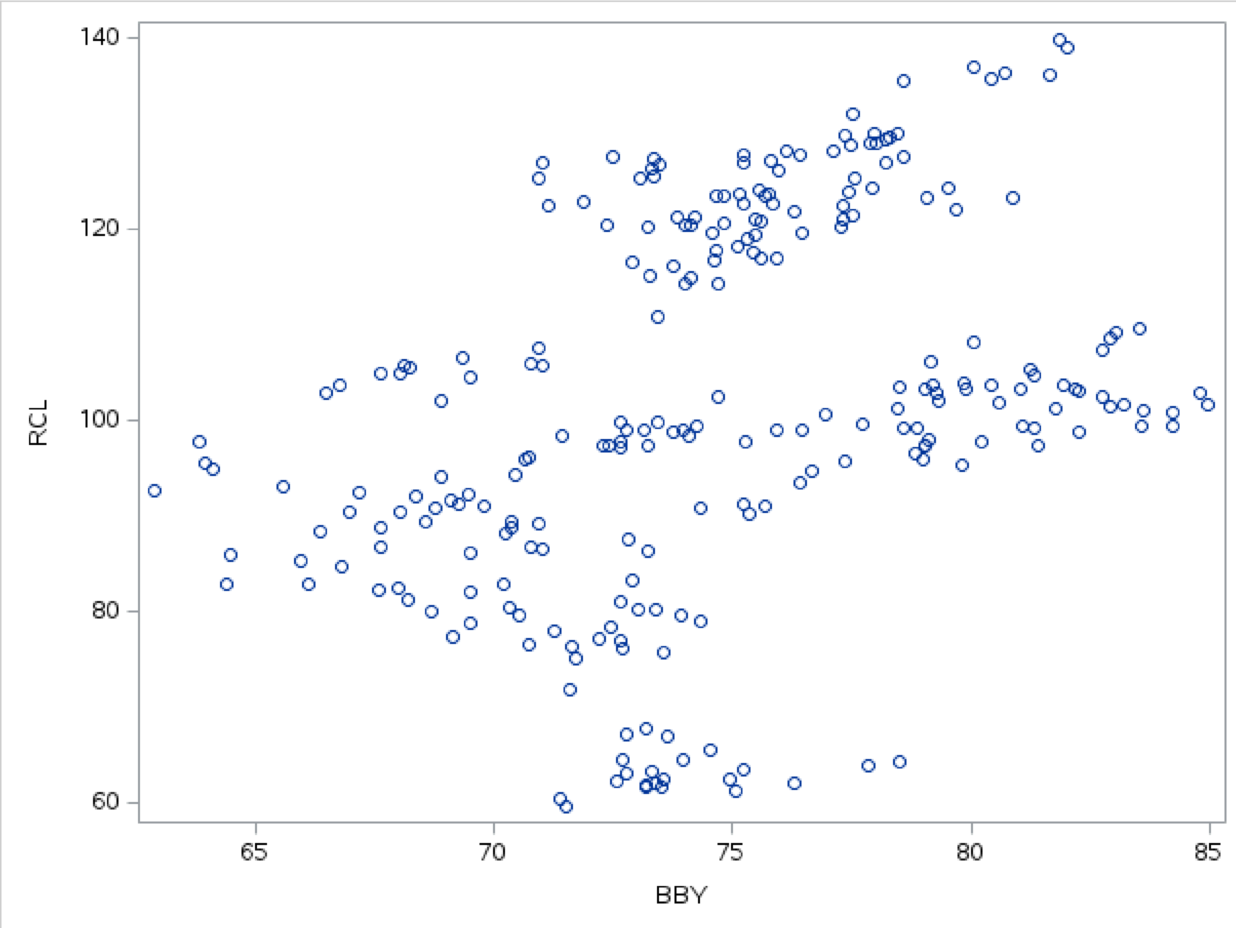
```
proc univariate data=prices plot;
var rcl;
run;
proc univariate data=prices plot;
var bby;
run;
proc corr data=prices;
var bby rcl;
run;
proc sgplot data=prices;
scatter x=bby y=rcl;
run;
proc univariate data=prices plot;
var bwa;
run;
proc corr data=prices;
var bwa rcl;
run;
proc sgplot data=prices;
scatter x=bwa y=rcl;
run;
proc univariate data=prices plot;
var ccl;
run;
proc corr data=prices;
var ccl rcl;
run;
proc sgplot data=prices;
scatter x=ccl y=rcl;
run;
proc univariate data=prices plot;
var ebay;
run;
proc corr data=prices;
var ebay rcl;
run;
proc univariate data=prices plot;
var expe;
run;
proc corr data=prices;
var expe rcl;
run;
proc univariate data=prices plot;
var grmn;
run;
proc corr data=prices;
var grmn rcl;
run;
proc sgplot data=prices;
scatter x=grmn y=rcl;
run;
proc univariate data=prices plot;
var has;
run;
proc corr data=prices;
var has rcl;
run;
proc sgplot data=prices;
scatter x=has y=rcl;
run;
```

```
proc univariate data=prices plot;
var kmx;
run;
proc corr data=prices;
var kmx rcl;
run;
proc sgplot data=prices;
scatter x=kmx y=rcl;
run;
proc univariate data=prices plot;
var low;
run;
proc corr data=prices;
var low rcl;
run;
proc sgplot data=prices;
scatter x=low y=rcl;
run;
proc univariate data=prices plot;
var nclh;
run;
proc corr data=prices;
var nclh rcl;
run;
proc sgplot data=prices;
scatter x=nclh y=rcl;
run;
```

```
proc univariate data=prices plot;
var phm;
run;
proc corr data=prices;
var phm rcl;
run;
proc sgplot data=prices;
scatter x=phm y=rcl;
run;
proc univariate data=prices plot;
var tsla;
run;
proc corr data=prices;
var tsla rcl;
run;
proc sgplot data=prices;
scatter x=tsla y=rcl;
run;
proc univariate data=prices plot;
var ulta;
run;
proc corr data=prices;
var ulta rcl;
run;
proc sgplot data=prices;
scatter x=ulta y=rcl;
run;
```

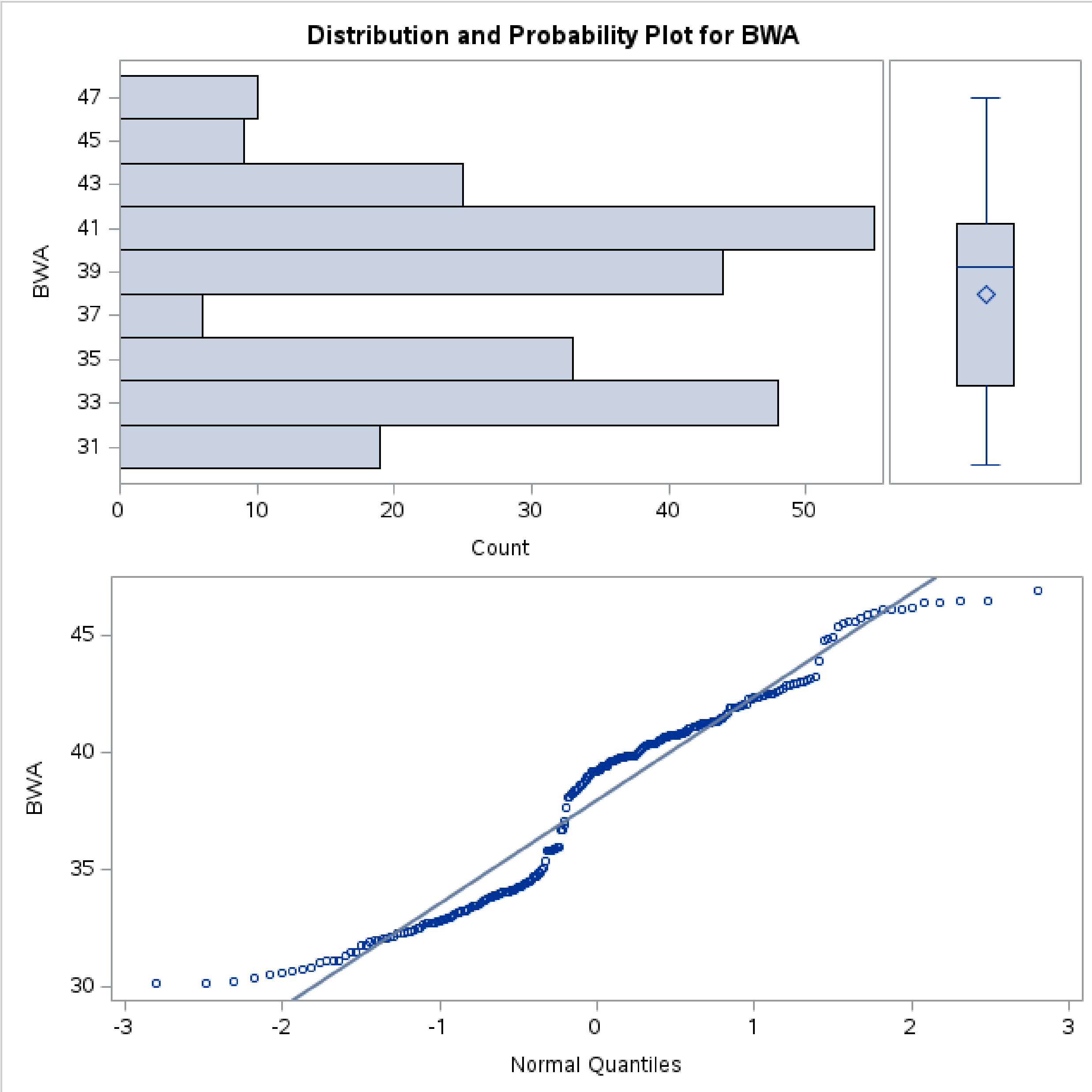

Distribution and Probability Plot for RCL
RCL
140
130
120
110
100
90
80
70
60
0
20
40
60
Count
RCL
140
120
100
80
60
-3
-2
-1
0
1
2
3
Normal Quantiles

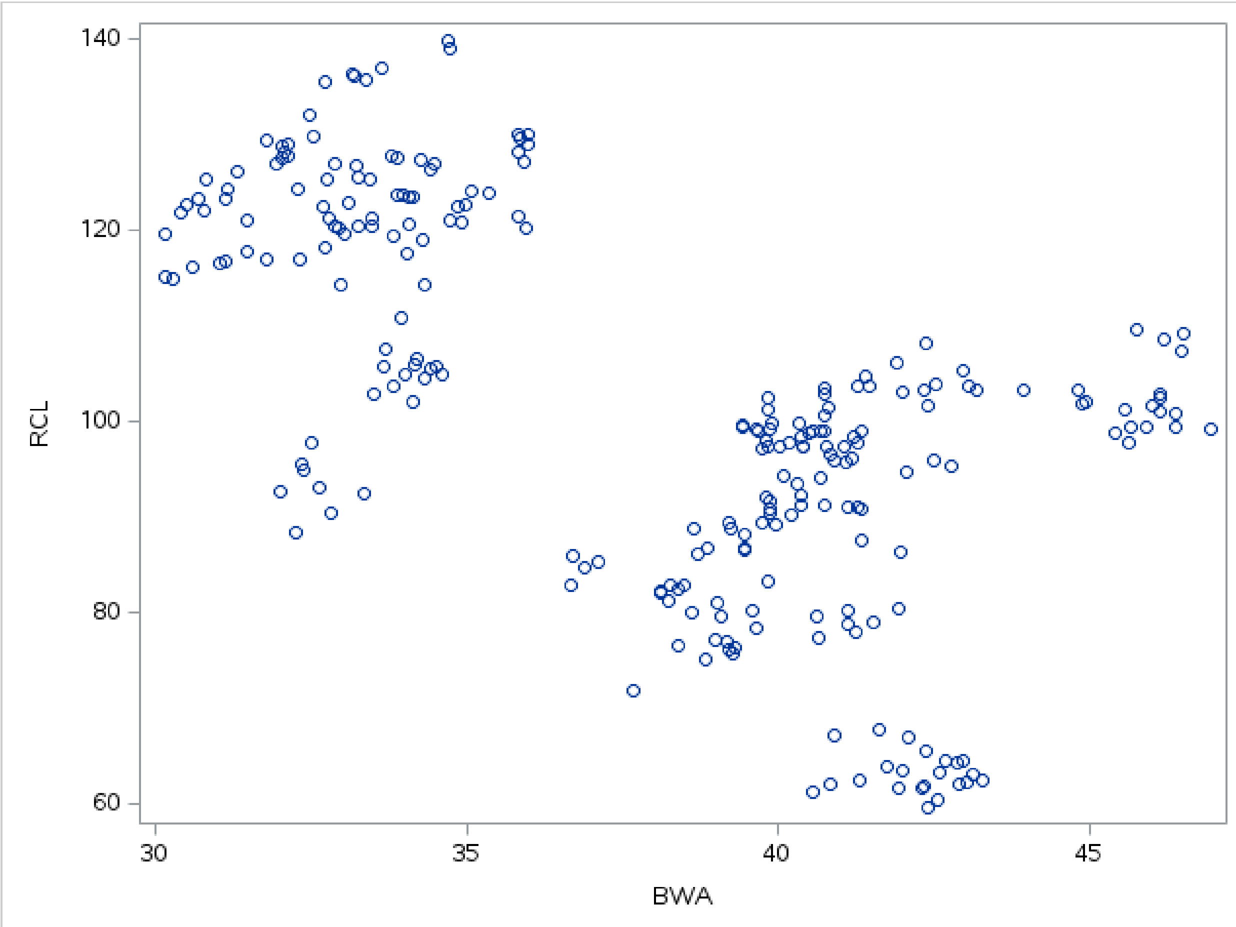

Distribution and Probability Plot for BBY
83.75
81.25
78.75
76.25
73.75
71.25
68.75
66.25
63.75
BBY
0
20
40
60
Count
85
80
75
70
65
BBY
-3
-2
-1
0
1
2
3
Normal Quantiles



BBYR

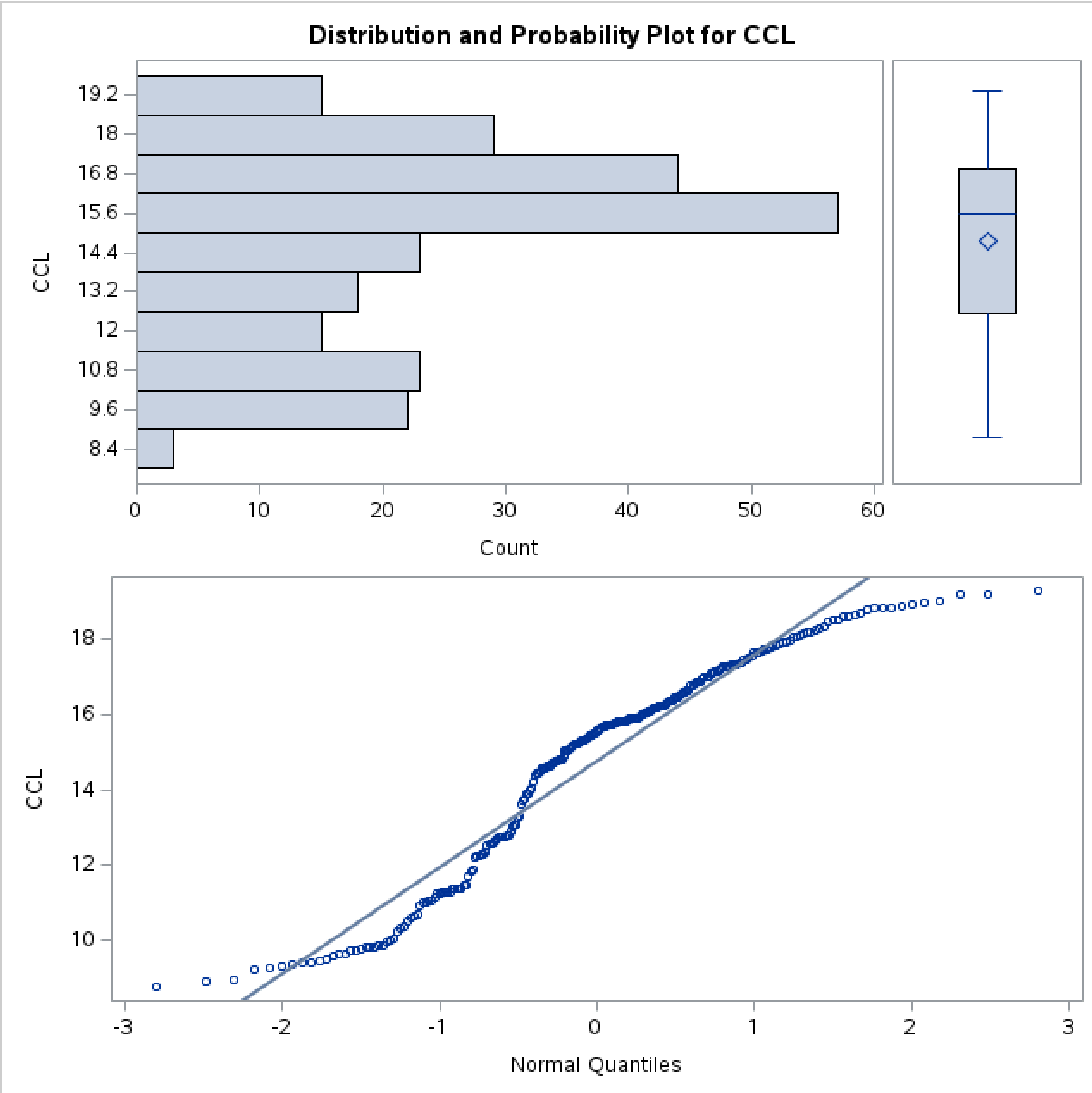
Pearson Correlation Coefficients, N = 249 Prob > |r| under H0: Rho=0		
	BBY	RCL
BBY BBY	1.00000	0.36345 <.0001
RCL RCL	0.36345 <.0001	1.00000

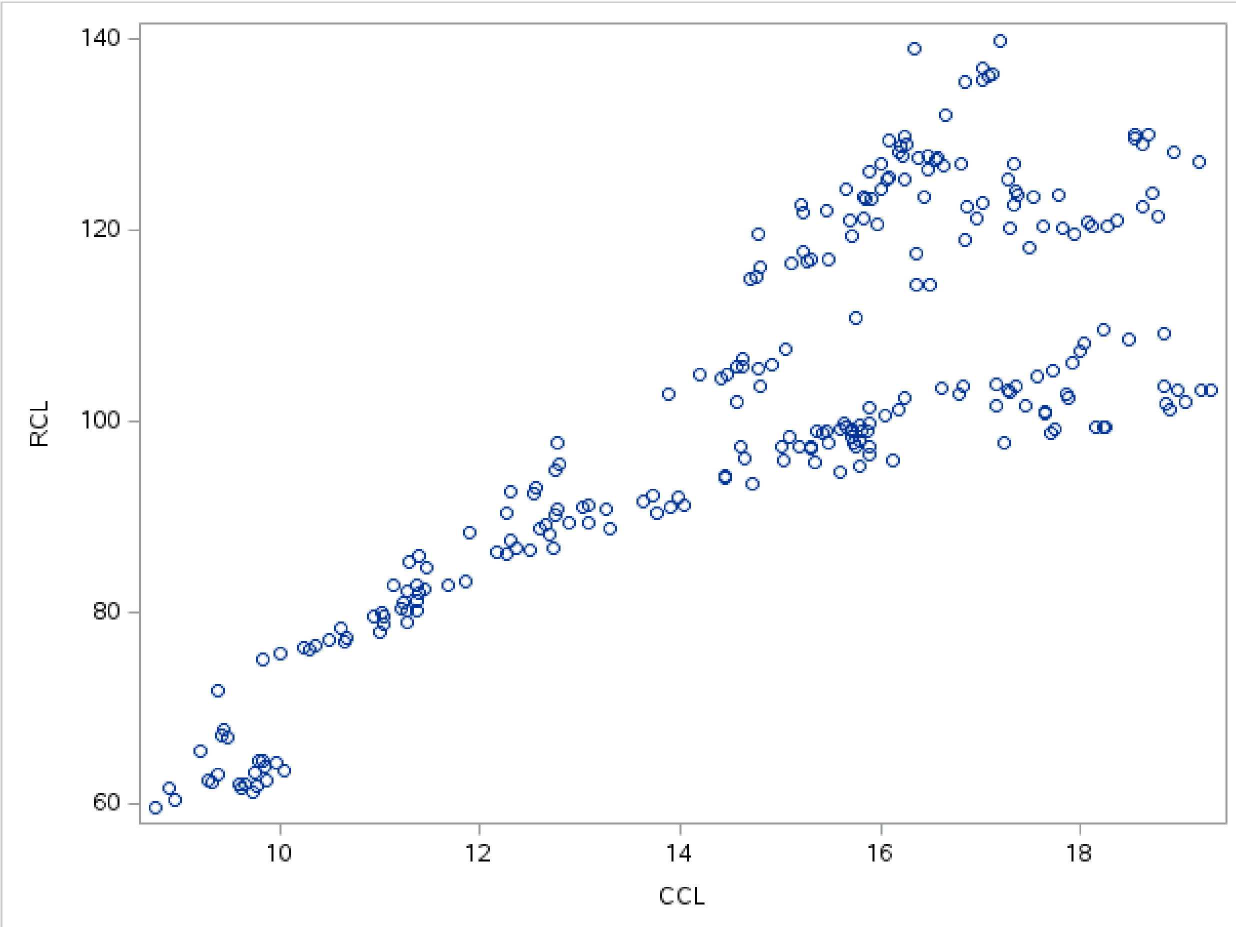

Distribution and Probability Plot for BWA
47
45
43
41
39
37
35
33
31
BWA
0
10
20
30
40
50
Count
45
40
35
30
BWA
-3
-2
-1
0
1
2
3
Normal Quantiles



BWAR

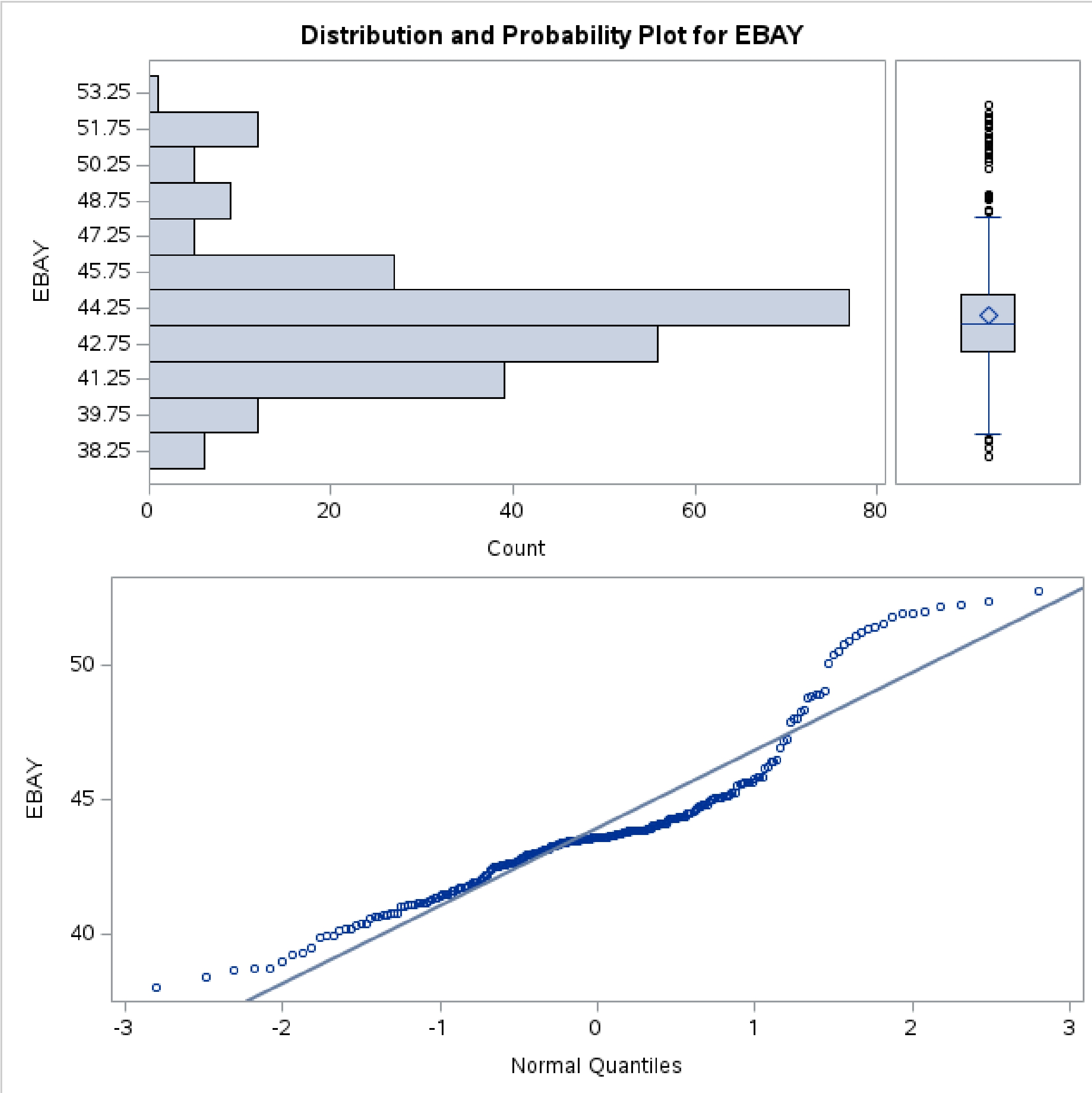
Pearson Correlation Coefficients, N = 249 Prob > |r| under H0: Rho=0		
	BWA	RCL
BWA BWA	1.00000	-0.61925 <.0001
RCL RCL	-0.61925 <.0001	1.00000

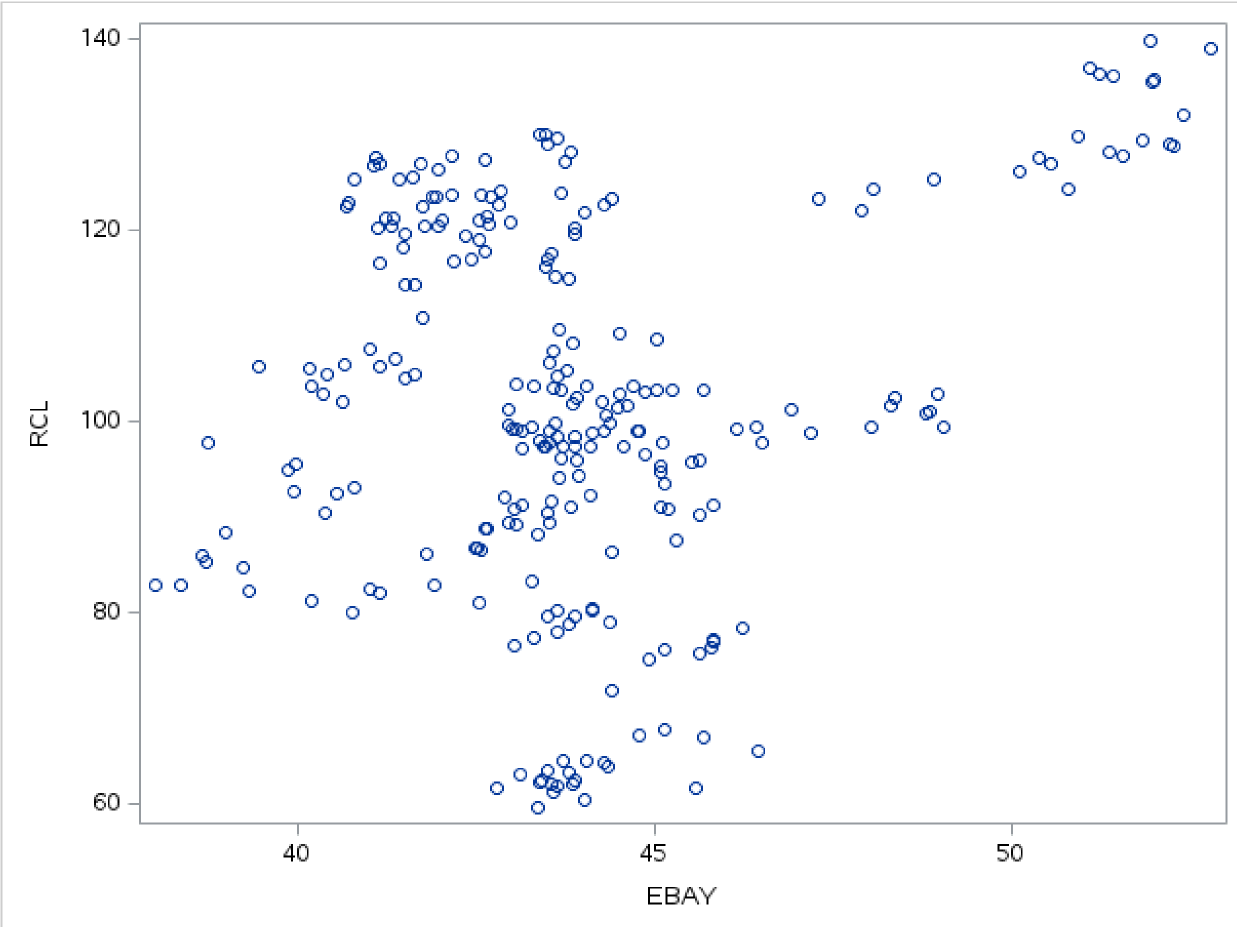

Distribution and Probability Plot for CCL
CCL
19.2
18
16.8
15.6
14.4
13.2
12
10.8
9.6
8.4
0
10
20
30
40
50
60
Count
CCL
18
16
14
12
10
-3
-2
-1
0
1
2
3
Normal Quantiles



CCLR

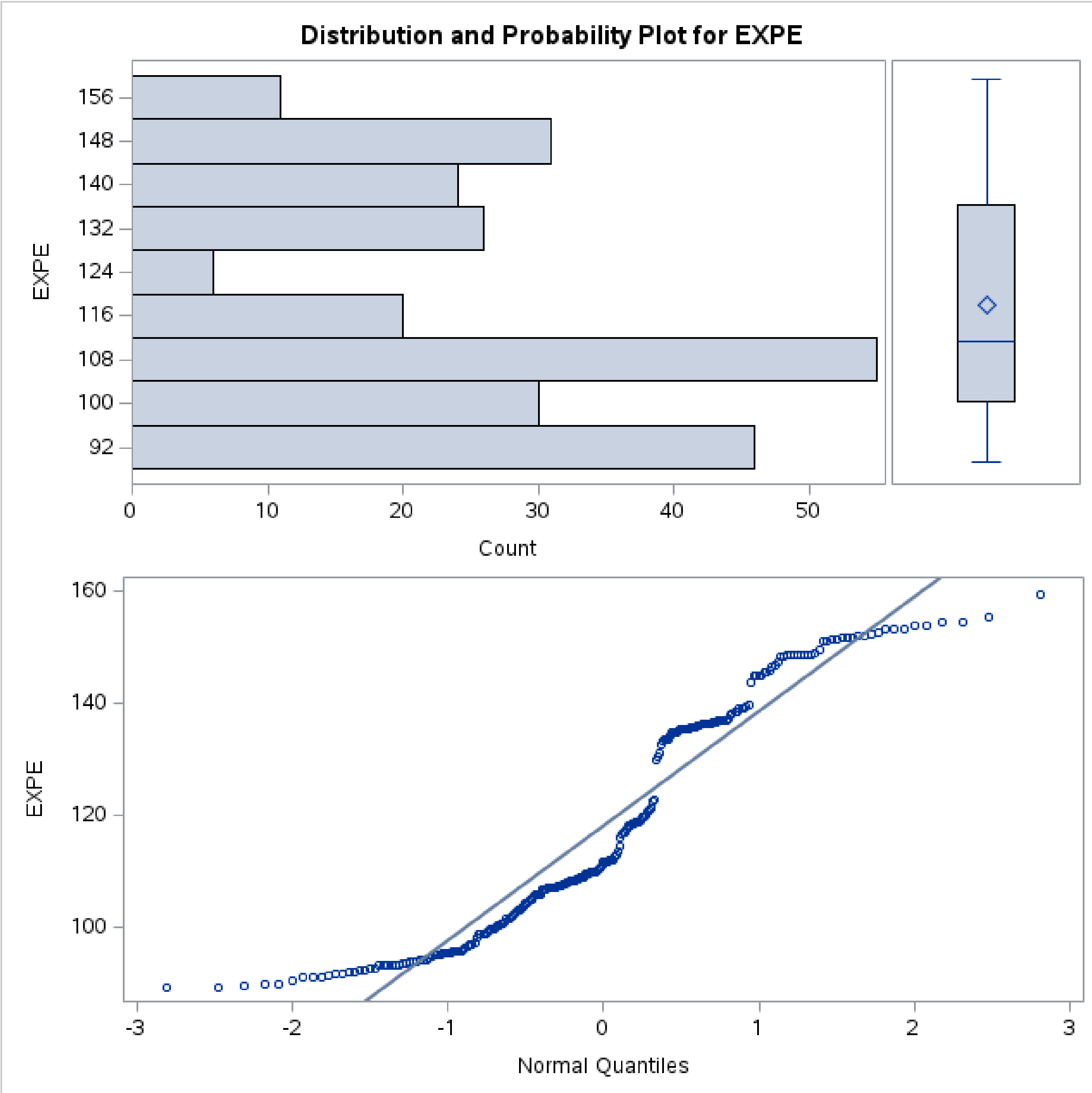
Pearson Correlation Coefficients, N = 249 Prob > |r| under H0: Rho=0		
	CCL	RCL
CCL CCL	1.00000	0.83042 <.0001
RCL RCL	0.83042 <.0001	1.00000

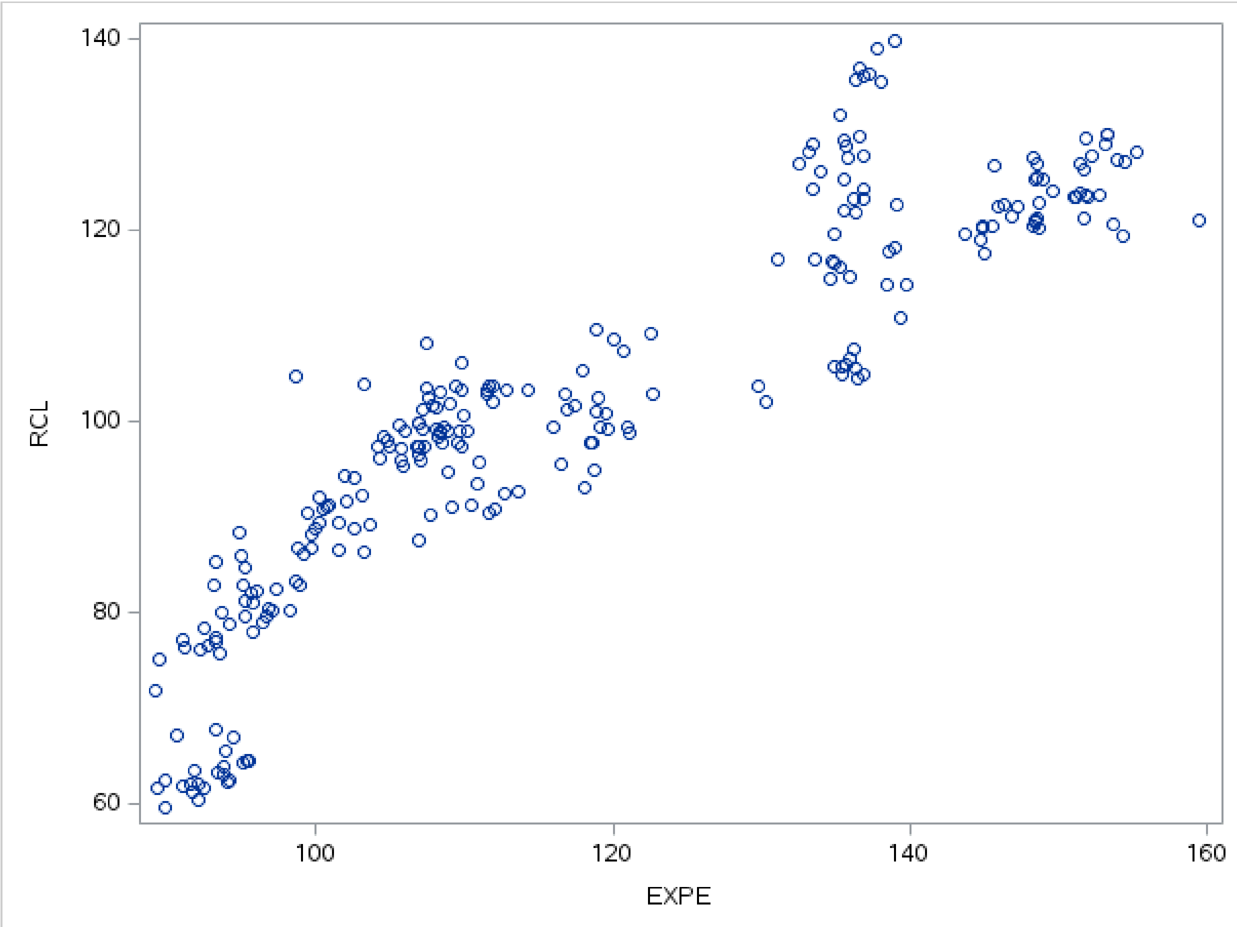

Distribution and Probability Plot for EBAY
EBAY
53.25
51.75
50.25
48.75
47.25
45.75
44.25
42.75
41.25
39.75
38.25
0
20
40
60
80
Count
EBAY
50
45
40
-3
-2
-1
0
1
2
3
Normal Quantiles



EBAYR

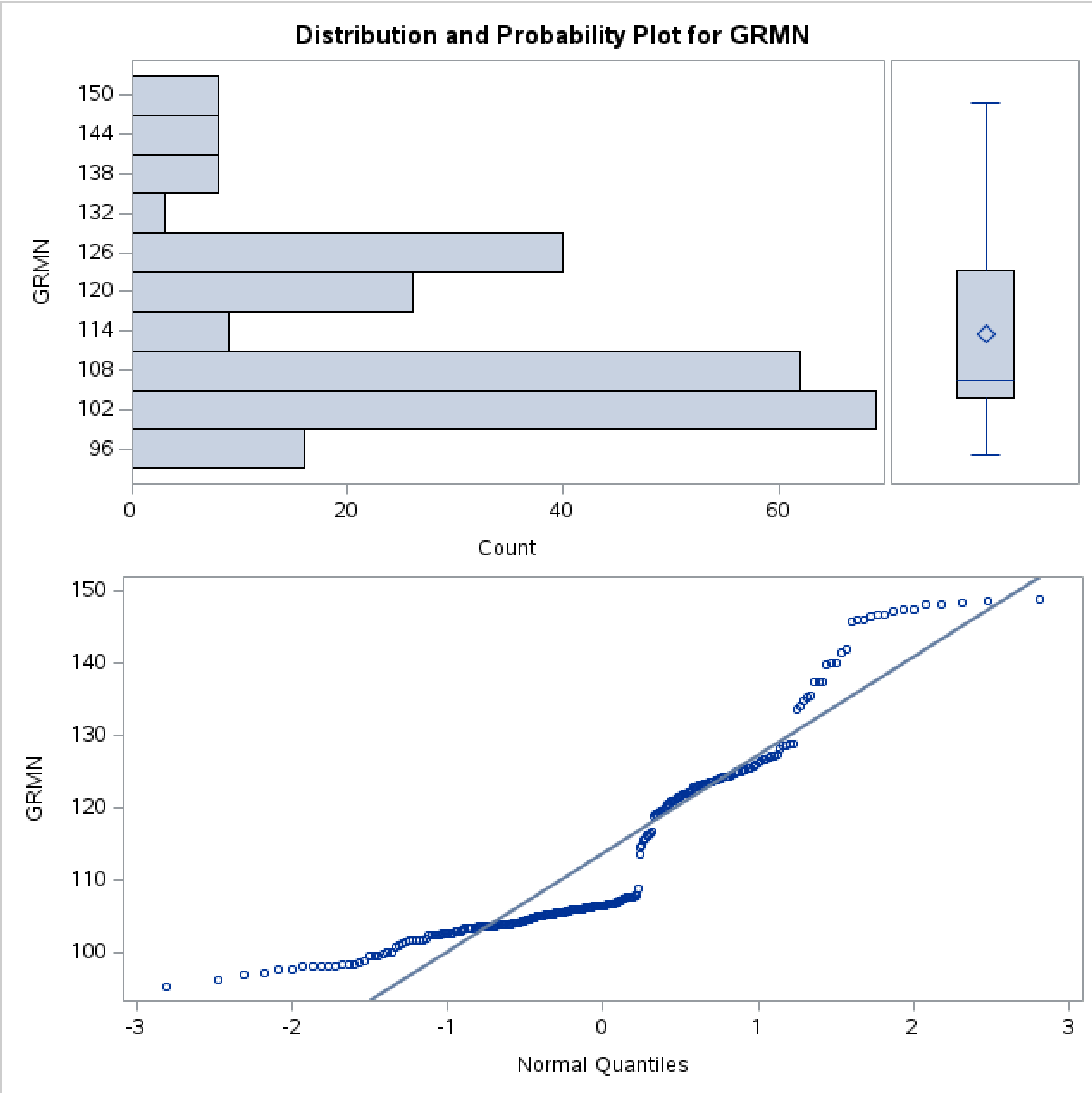
Pearson Correlation Coefficients, N = 249 Prob > |r| under H0: Rho=0		
	EBAY	RCL
EBAY EBAY	1.00000	0.23736 0.0002
RCL RCL	0.23736 0.0002	1.00000

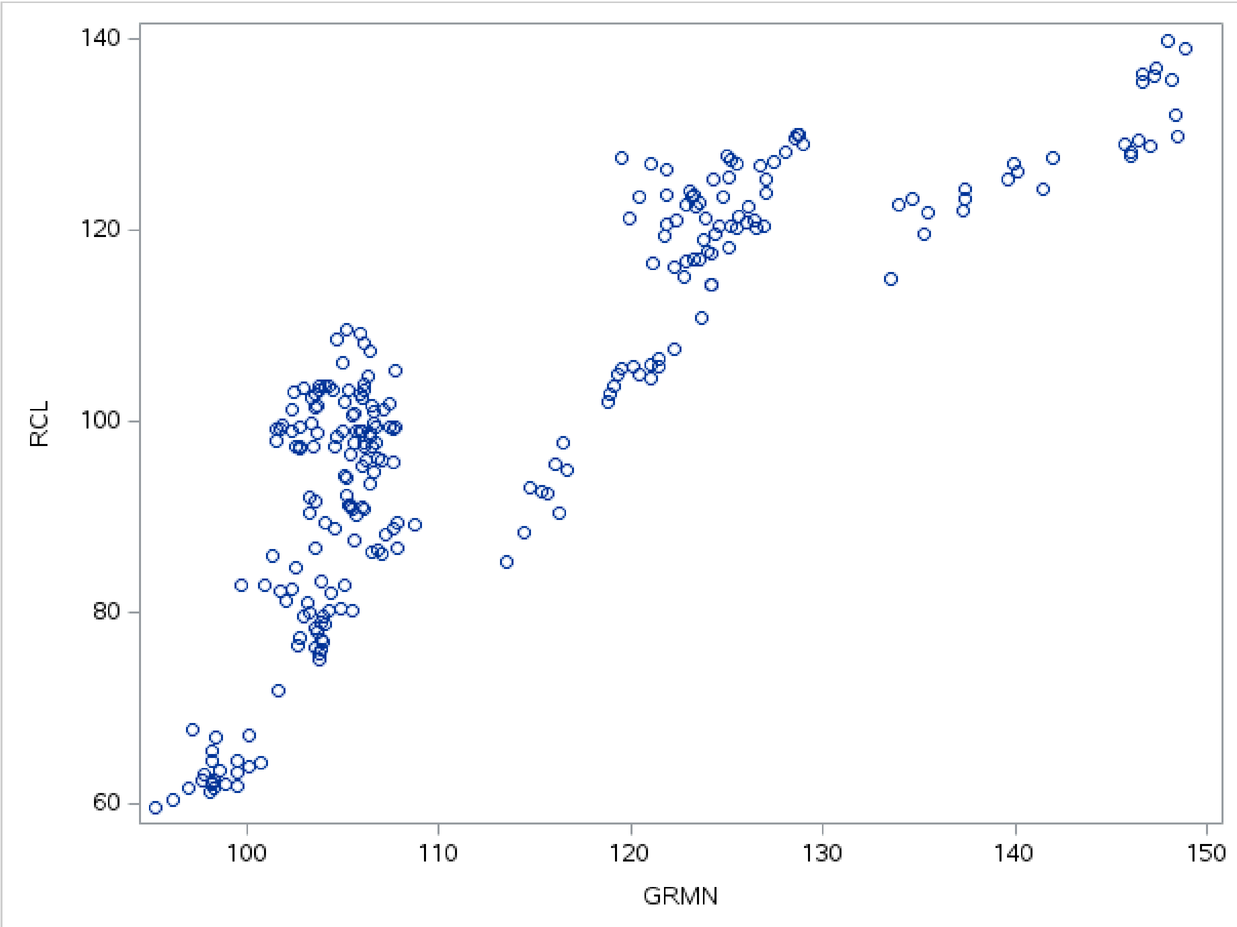

Distribution and Probability Plot for EXPE
EXPE
156
148
140
132
124
116
108
100
92
0
10
20
30
40
50
Count
160
140
120
100
-3
-2
-1
0
1
2
3
Normal Quantiles



EXPER

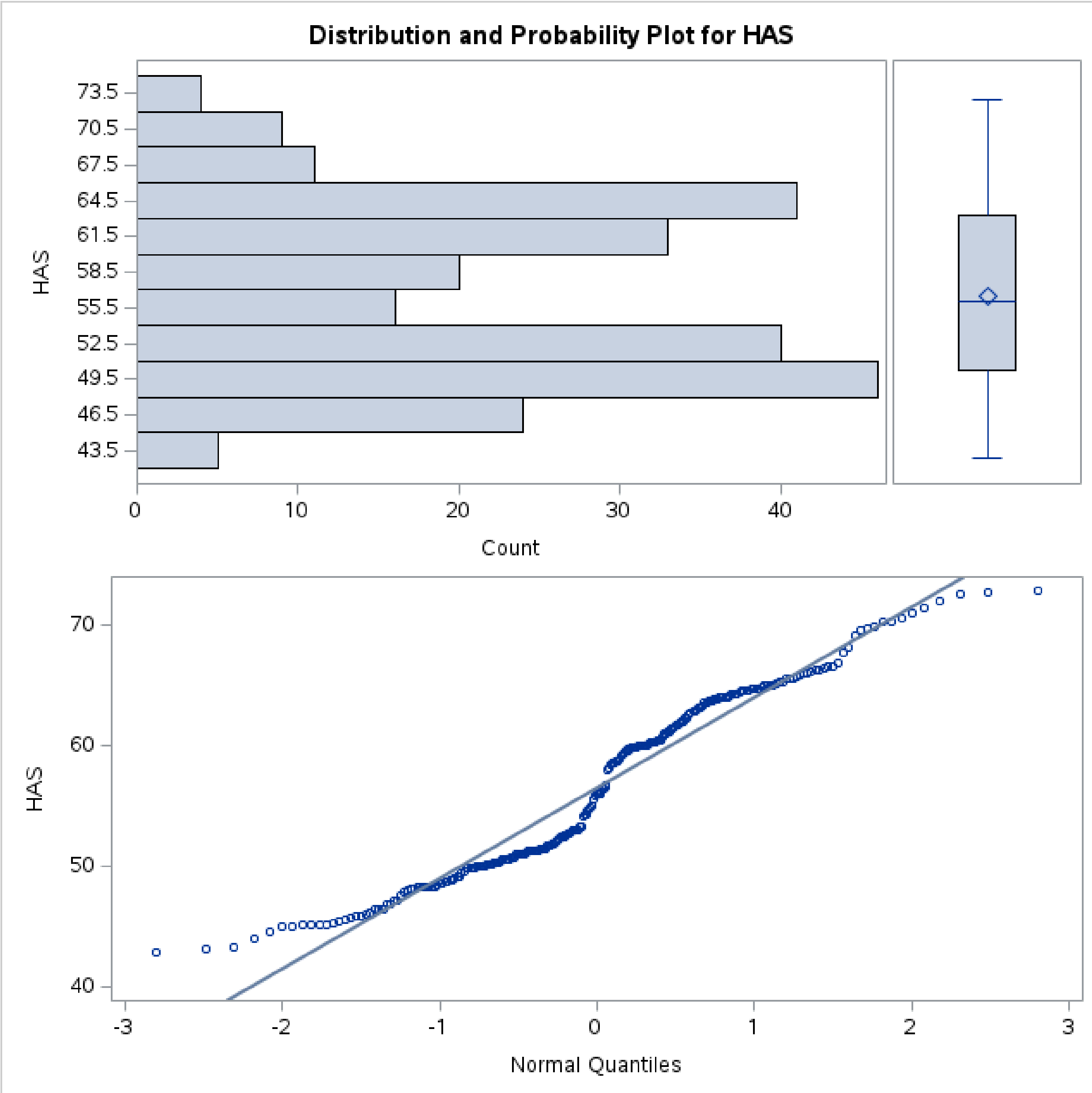
Pearson Correlation Coefficients, N = 249 Prob > |r| under H0: Rho=0		
	EXPE	RCL
EXPE EXPE	1.00000	0.90311 <.0001
RCL RCL	0.90311 <.0001	1.00000

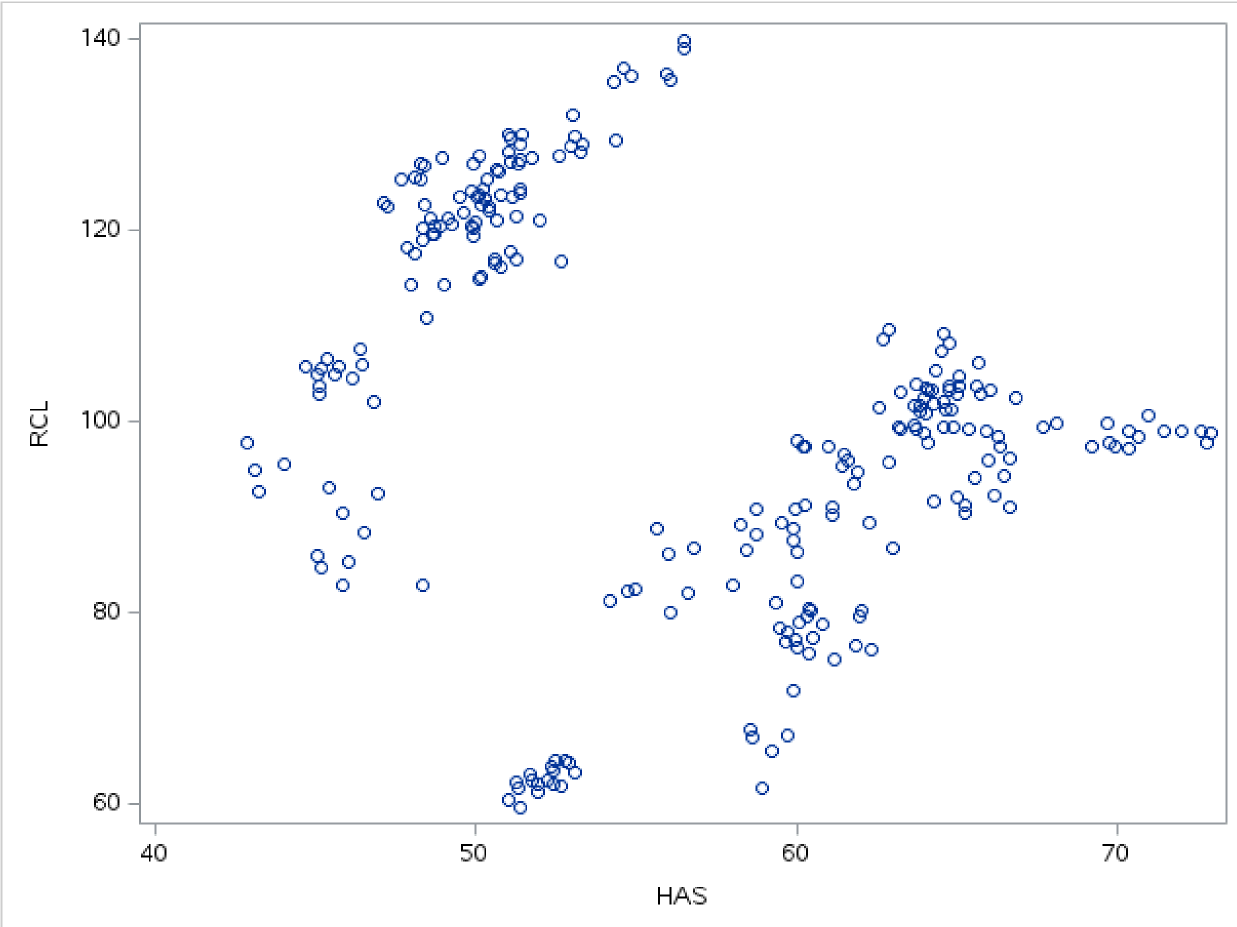

Distribution and Probability Plot for GRMN
GRMN
150
144
138
132
126
120
114
108
102
96
0
20
40
60
Count
GRMN
150
140
130
120
110
100
-3
-2
-1
0
1
2
3
Normal Quantiles



GRMNR

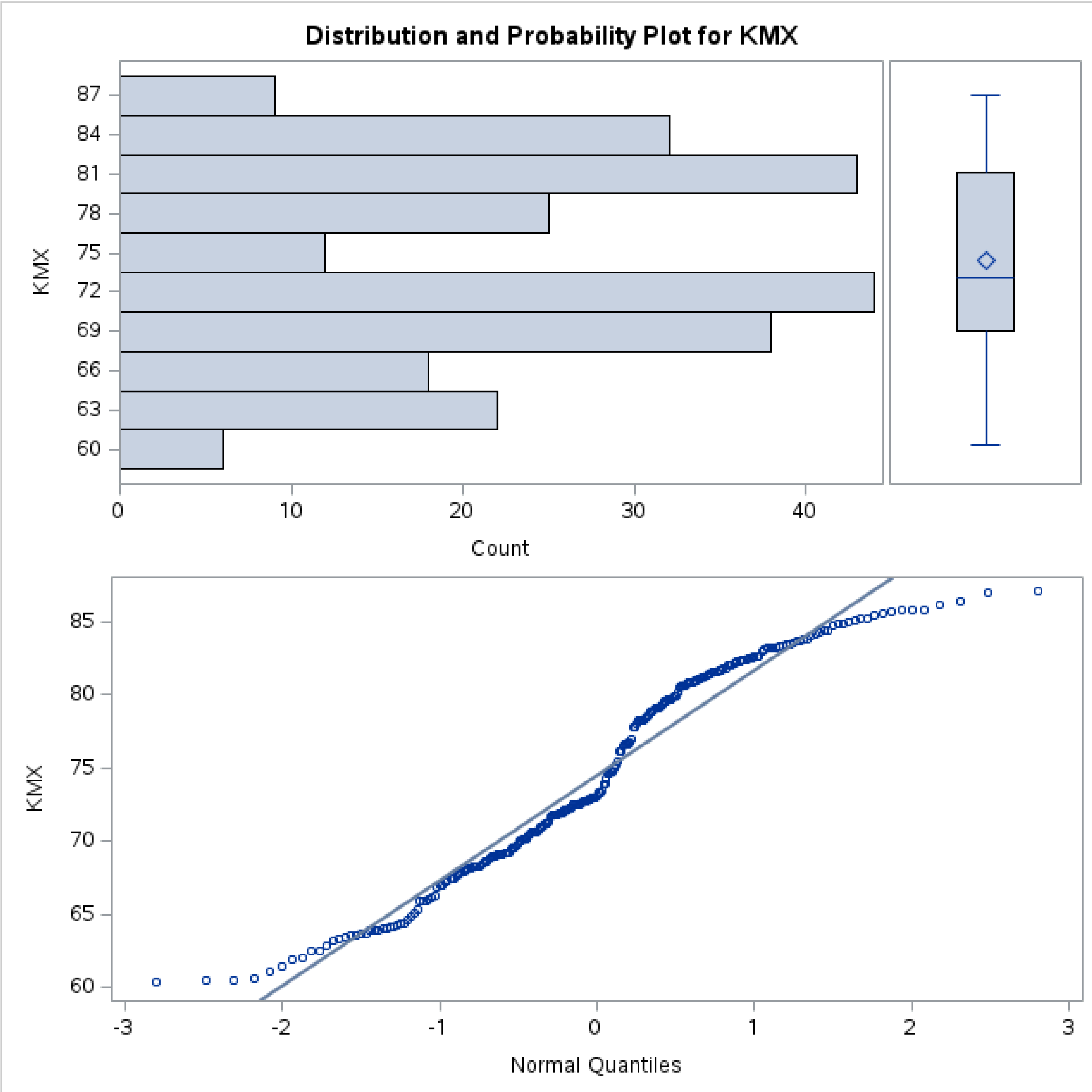
Pearson Correlation Coefficients, N = 249 Prob > |r| under H0: Rho=0		
	GRMN	RCL
GRMN GRMN	1.00000	0.84953 <.0001
RCL RCL	0.84953 <.0001	1.00000


Distribution and Probability Plot for HAS
HAS
73.5
70.5
67.5
64.5
61.5
58.5
55.5
52.5
49.5
46.5
43.5
0
10
20
30
40
Count
HAS
70
60
50
40
-3
-2
-1
0
1
2
3
Normal Quantiles

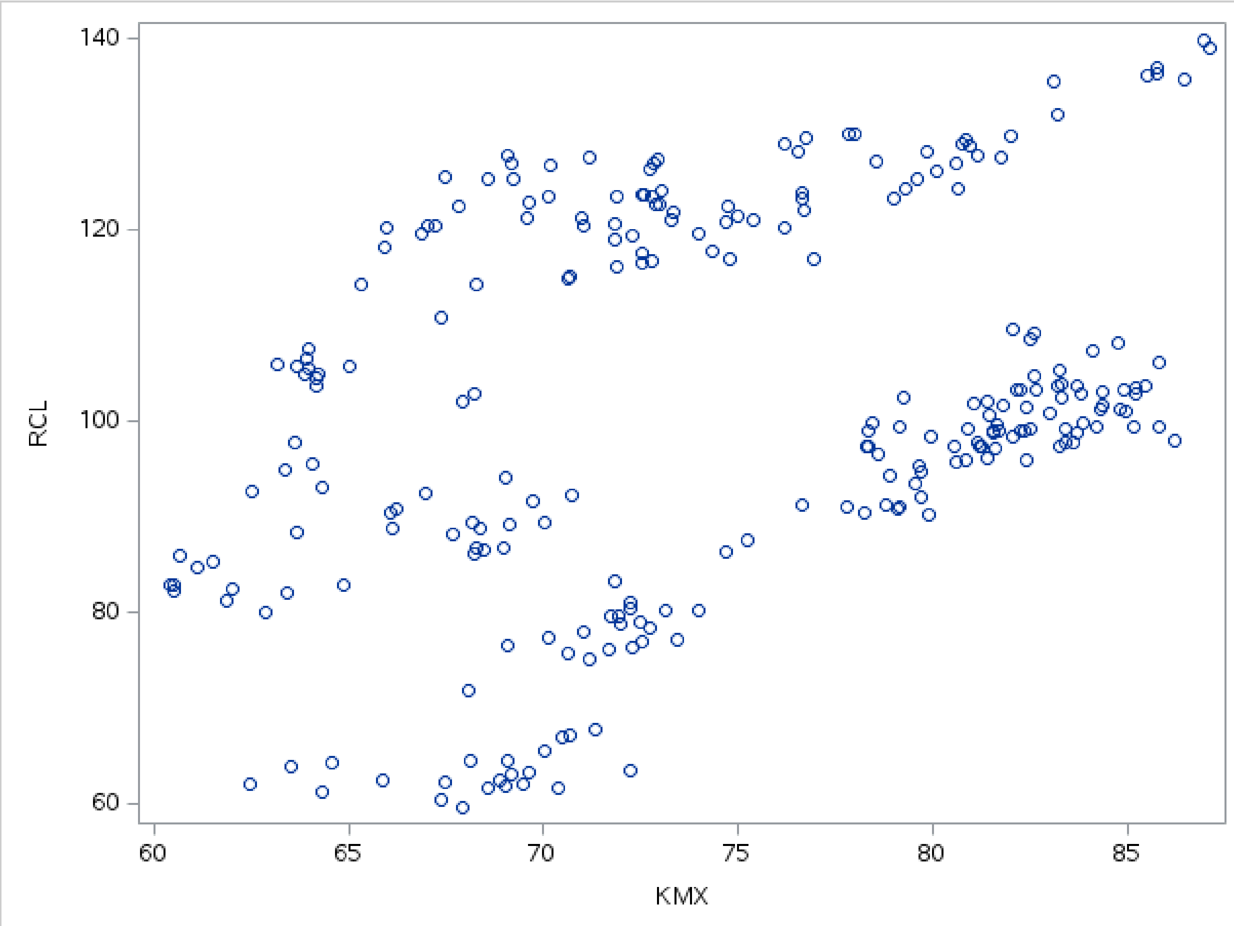


HASR

Pearson Correlation Coefficients, N = 249 Prob > |r| under H0: Rho=0		
	HAS	**RCL**
HAS **HAS**	1.00000	-0.27368 <.0001
RCL **RCL**	-0.27368 <.0001	1.00000


Distribution and Probability Plot for KMX
KMX
87
84
81
78
75
72
69
66
63
60
0
10
20
30
40
Count
85
80
75
70
65
60
-3
-2
-1
0
1
2
3
Normal Quantiles

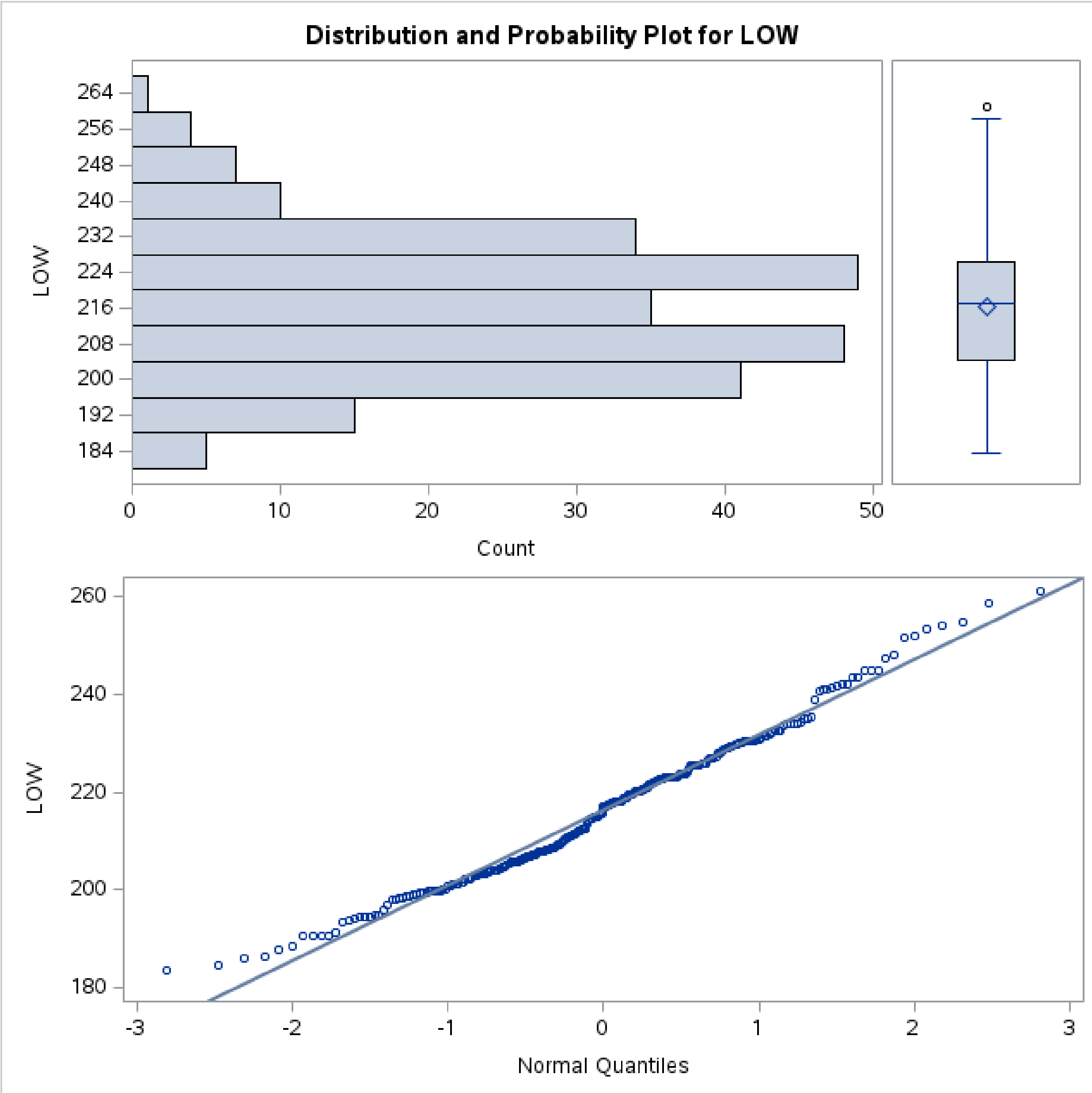
33

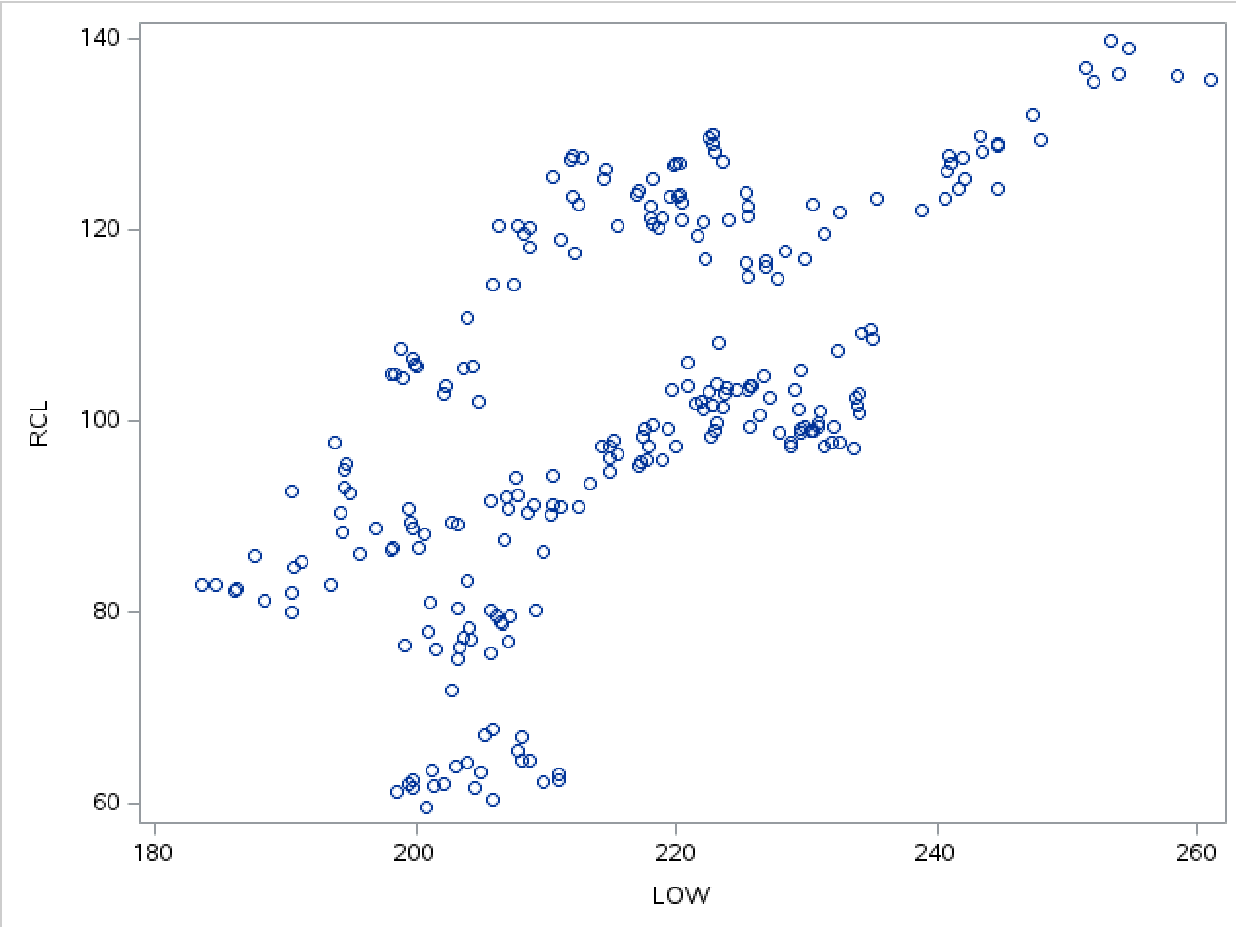


KMXR

Pearson Correlation Coefficients, N = 249 Prob > |r| under H0: Rho=0		
	KMX	RCL
KMX KMX	1.00000	0.35104 <.0001
RCL RCL	0.35104 <.0001	1.00000

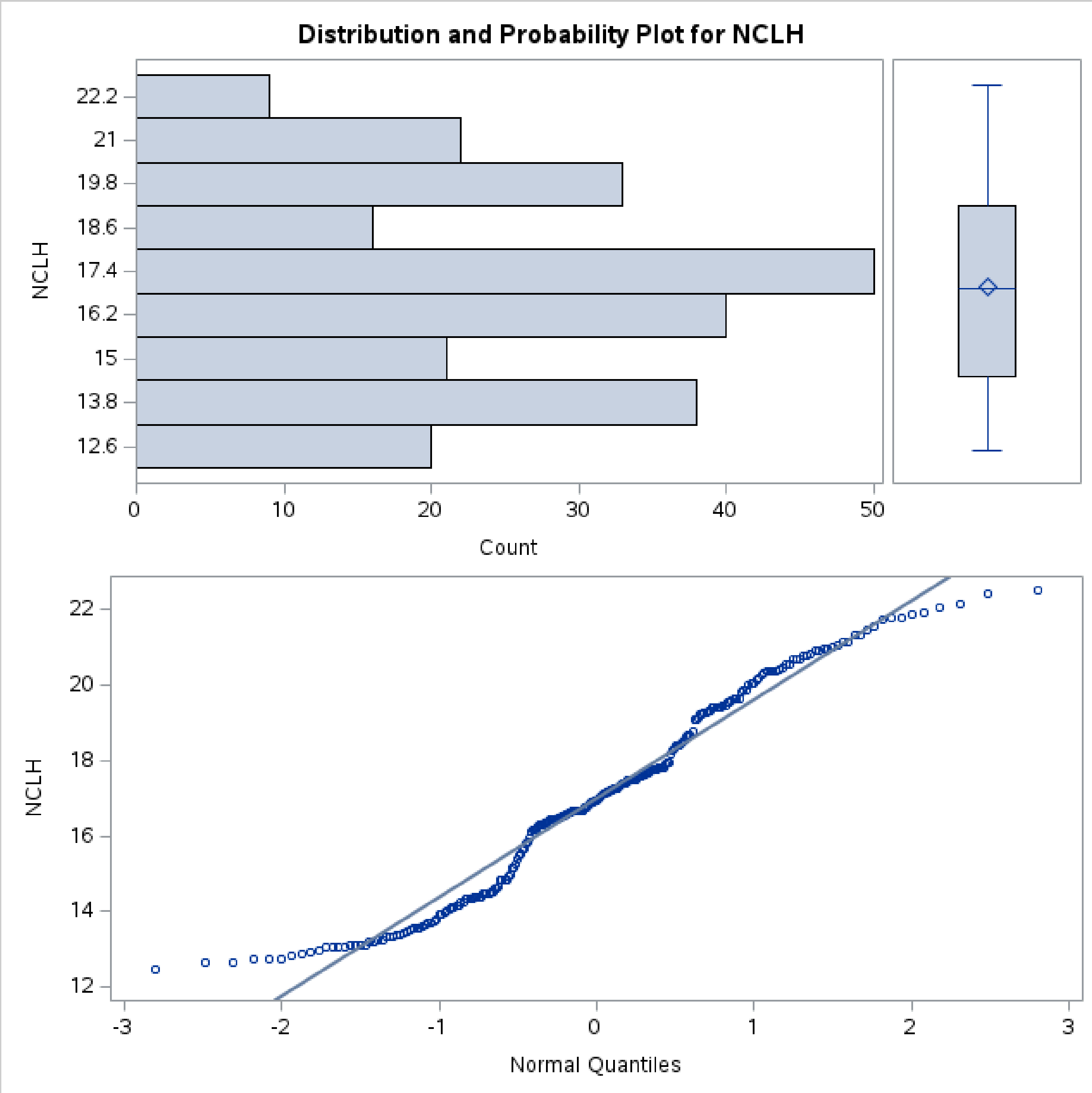
34

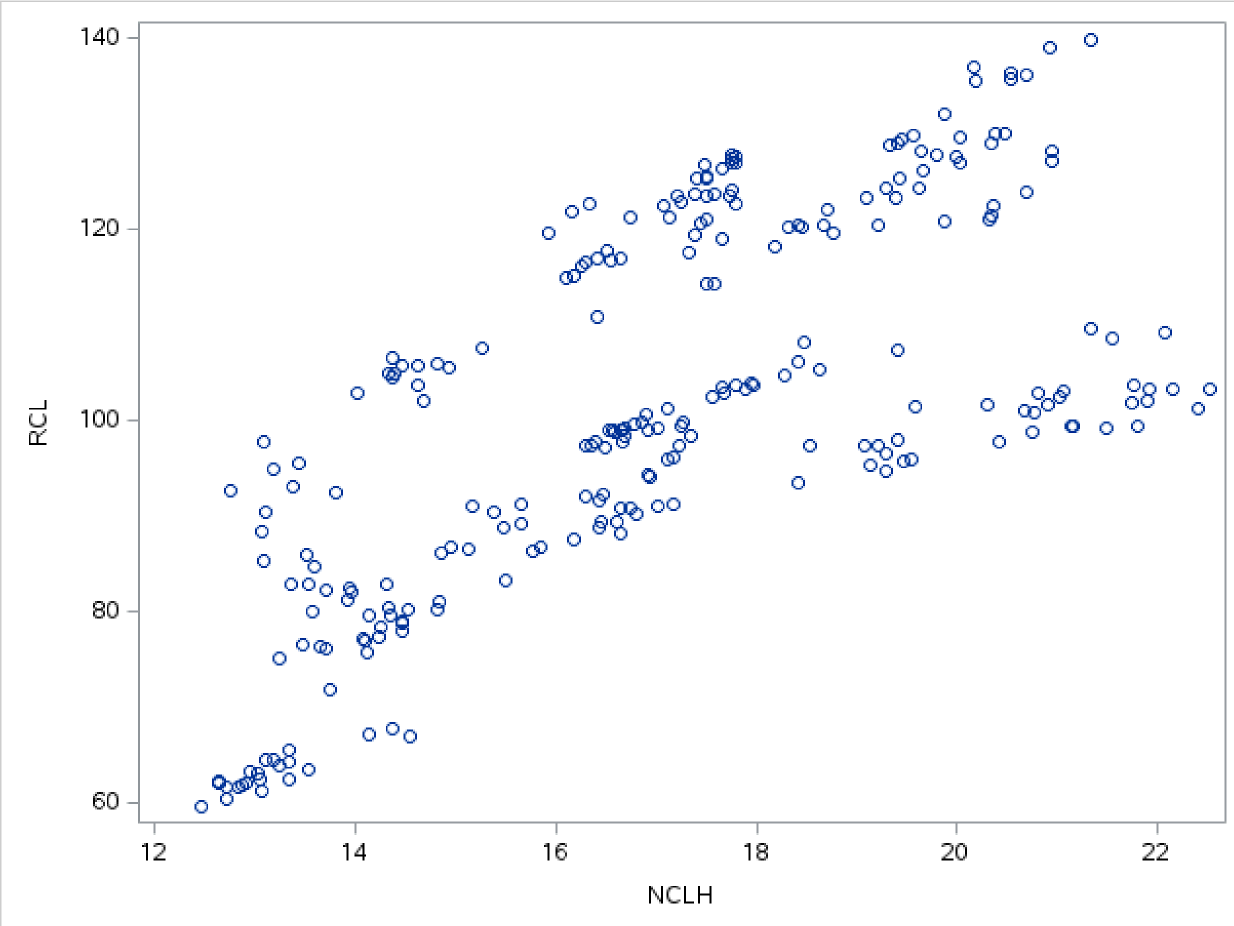

Distribution and Probability Plot for LOW
LOW
264
256
248
240
232
224
216
208
200
192
184
0
10
20
30
40
50
Count
LOW
260
240
220
200
180
-3
-2
-1
0
1
2
3
Normal Quantiles



LOWR

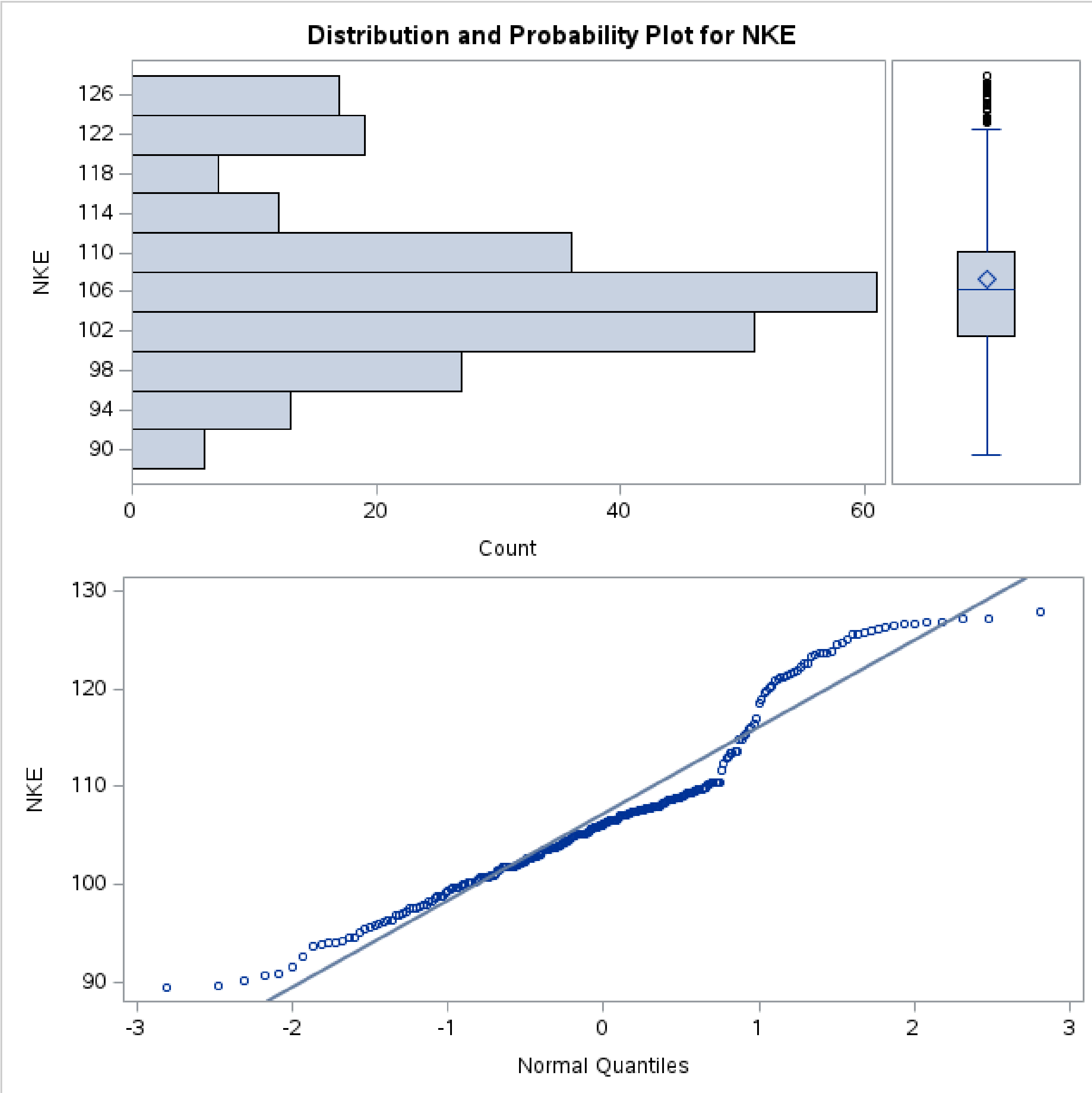
Pearson Correlation Coefficients, N = 249 Prob > |r| under H0: Rho=0		
	LOW	RCL
LOW LOW	1.00000	0.65013 <.0001
RCL RCL	0.65013 <.0001	1.00000

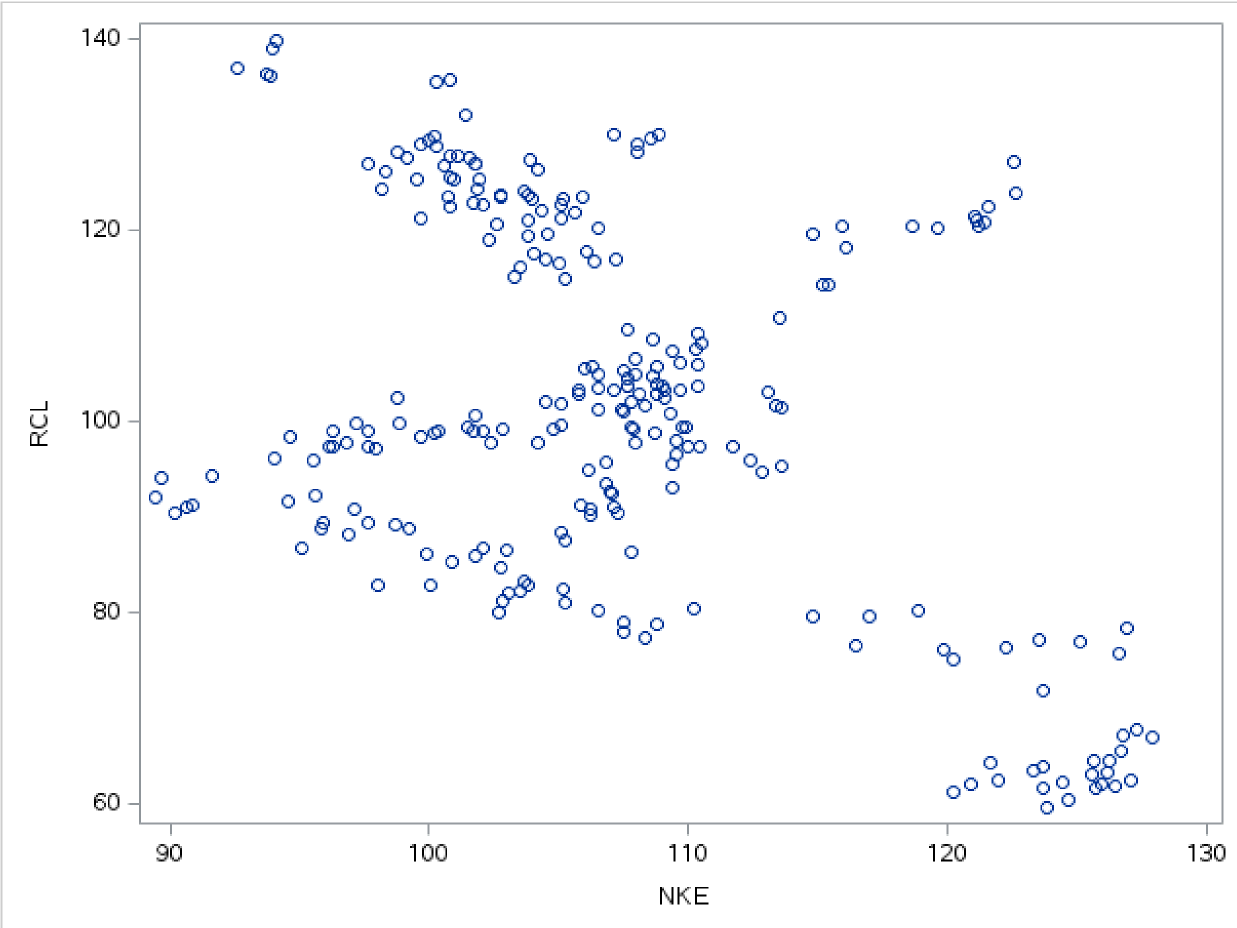

Distribution and Probability Plot for NCLH
NCLH
22.2
21
19.8
18.6
17.4
16.2
15
13.8
12.6
0
10
20
30
40
50
Count
NCLH
22
20
18
16
14
12
-3
-2
-1
0
1
2
3
Normal Quantiles



NCLHR

Pearson Correlation Coefficients, N = 249 Prob > |r| under H0: Rho=0		
	NCLH	RCL
NCLH NCLH	1.00000	0.69455 <.0001
RCL RCL	0.69455 <.0001	1.00000

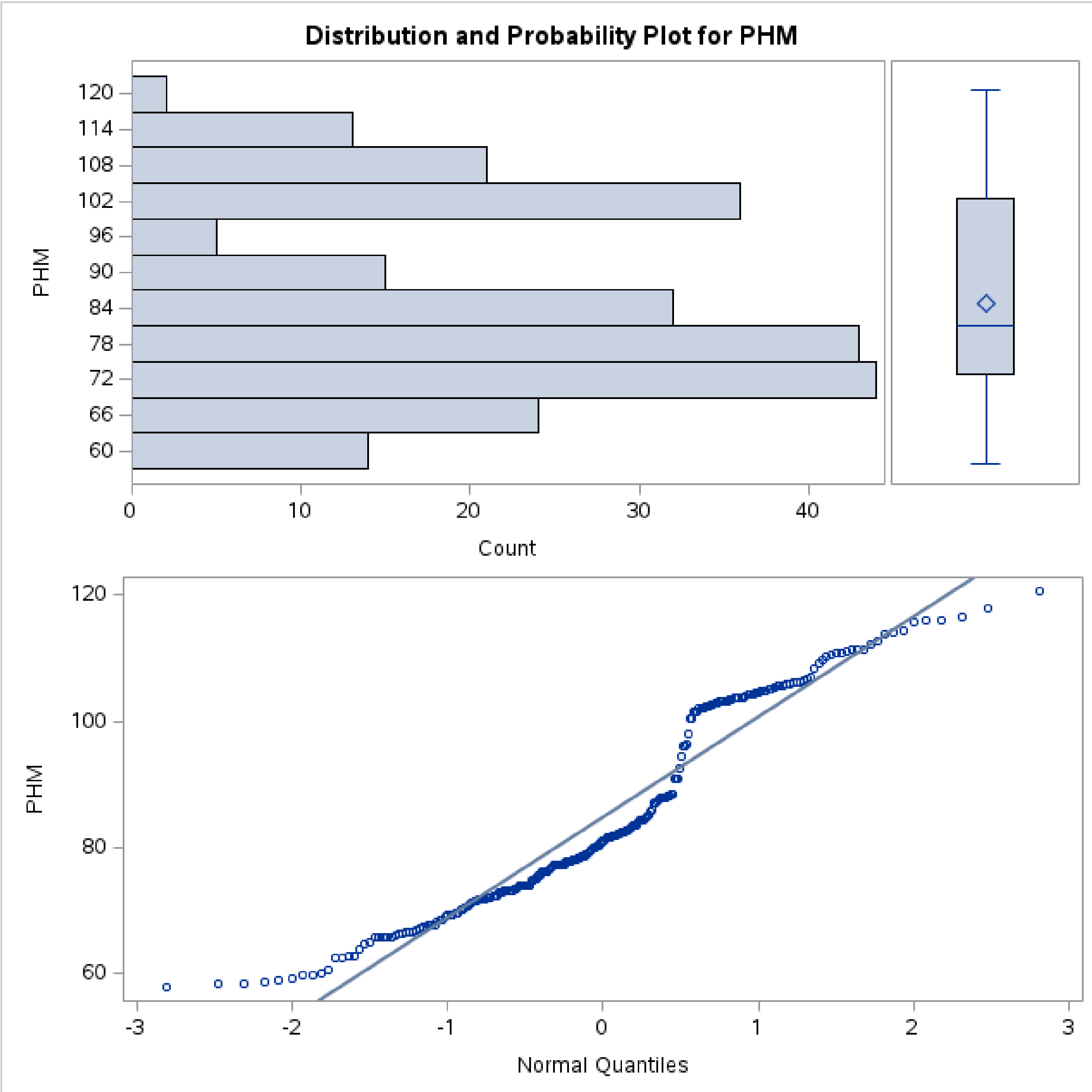

Distribution and Probability Plot for NKE
NKE
126
122
118
114
110
106
102
98
94
90
0
20
40
60
Count
130
120
110
100
90
-3
-2
-1
0
1
2
3
Normal Quantiles



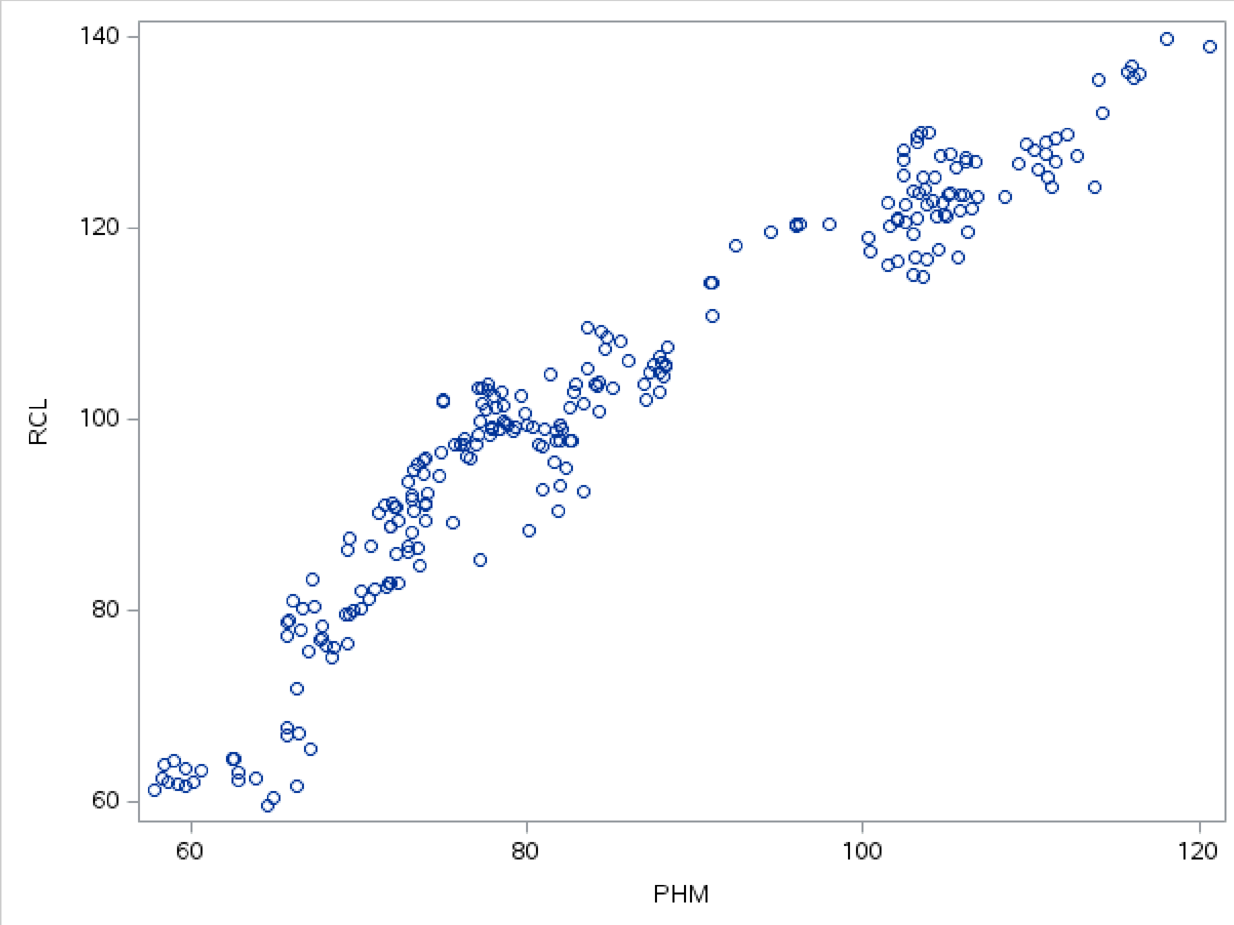
RNKE

Pearson Correlation Coefficients, N = 249 Prob > |r| under H0: Rho=0		
	NKE	RCL
NKE NKE	1.00000	-0.46382 <.0001
RCL RCL	-0.46382 <.0001	1.00000

40

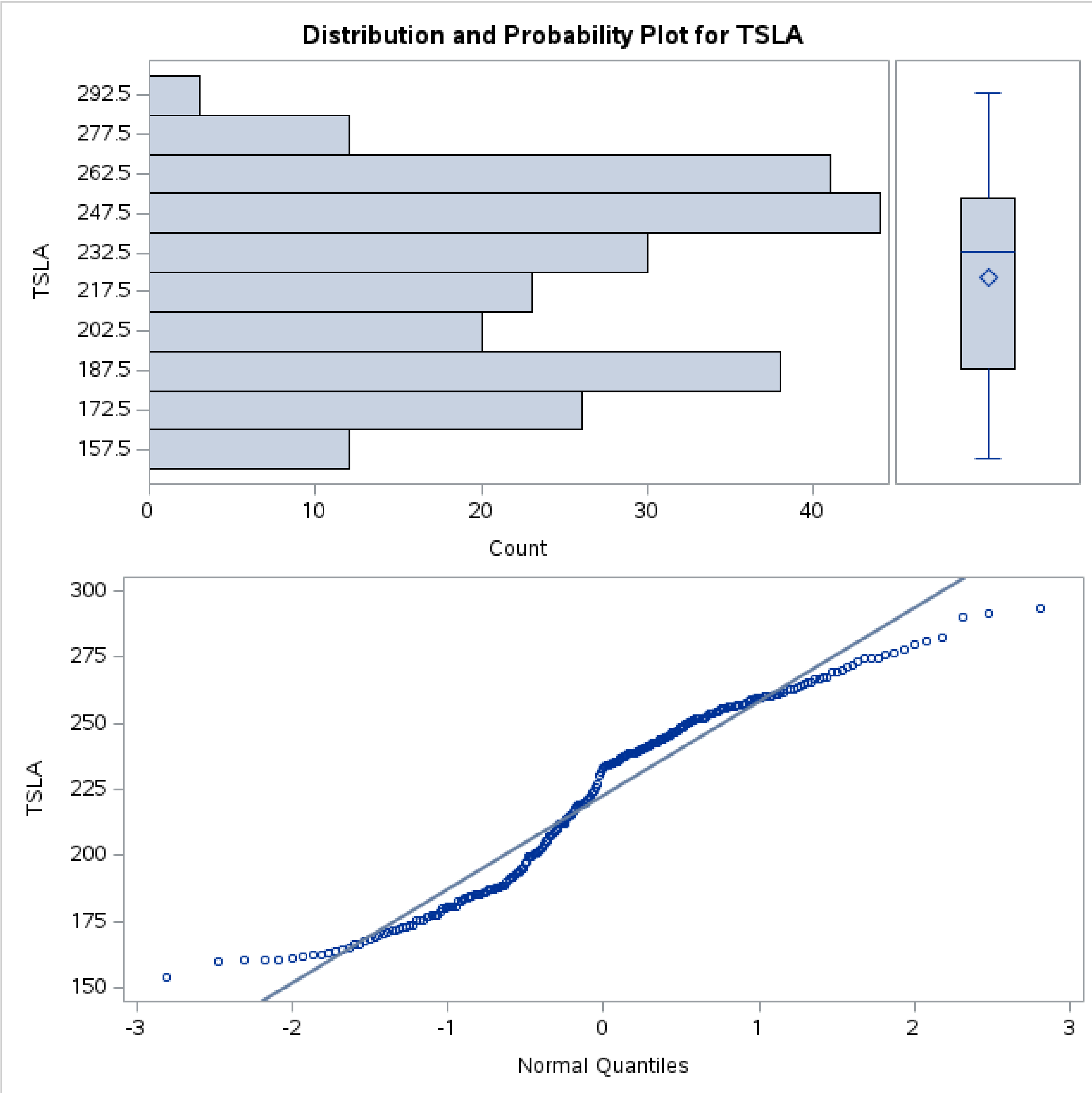

Distribution and Probability Plot for PHM
PHM
120
114
108
102
96
90
84
78
72
66
60
0
10
20
30
40
Count
120
100
80
60
-3
-2
-1
0
1
2
3
Normal Quantiles

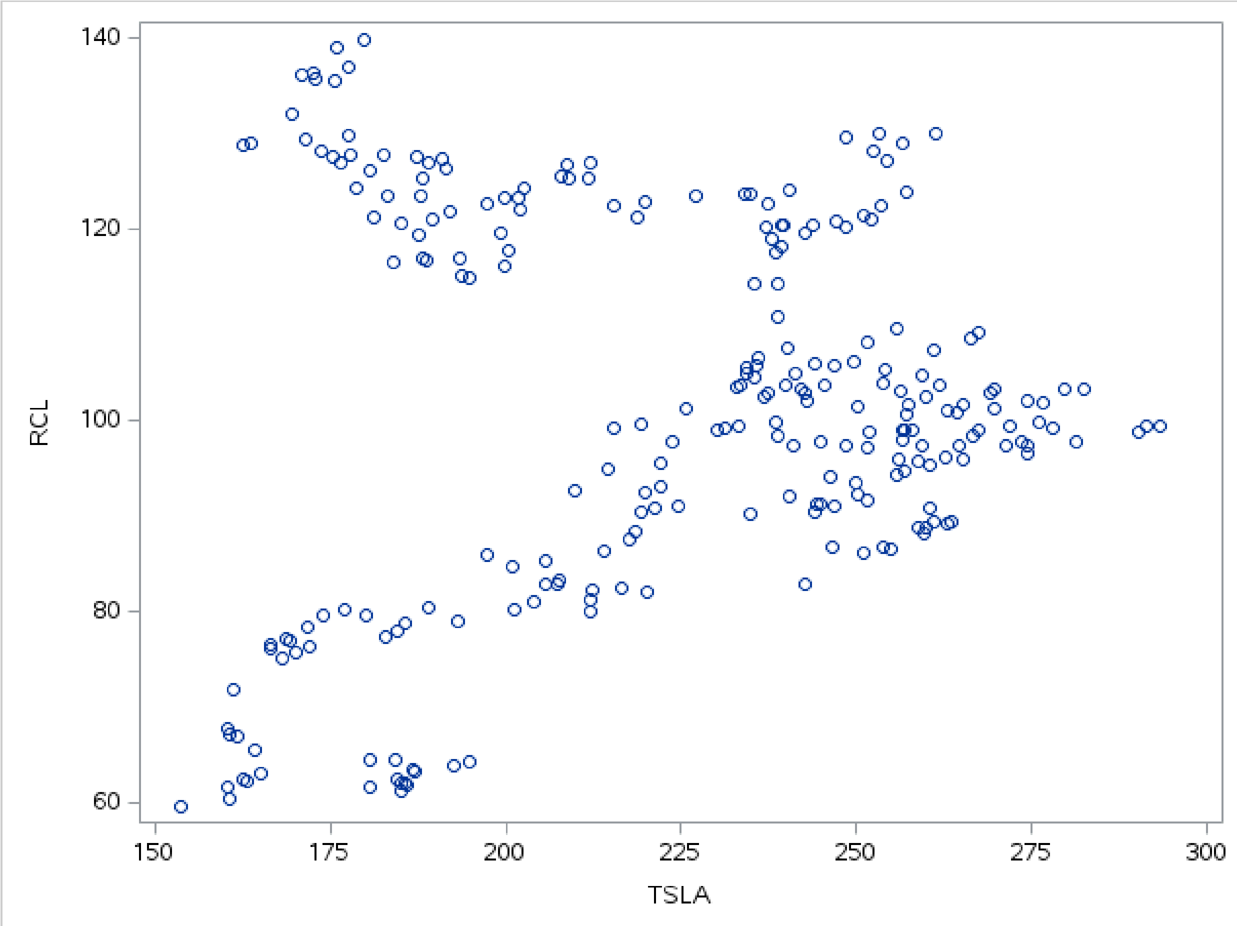
41



PHMR

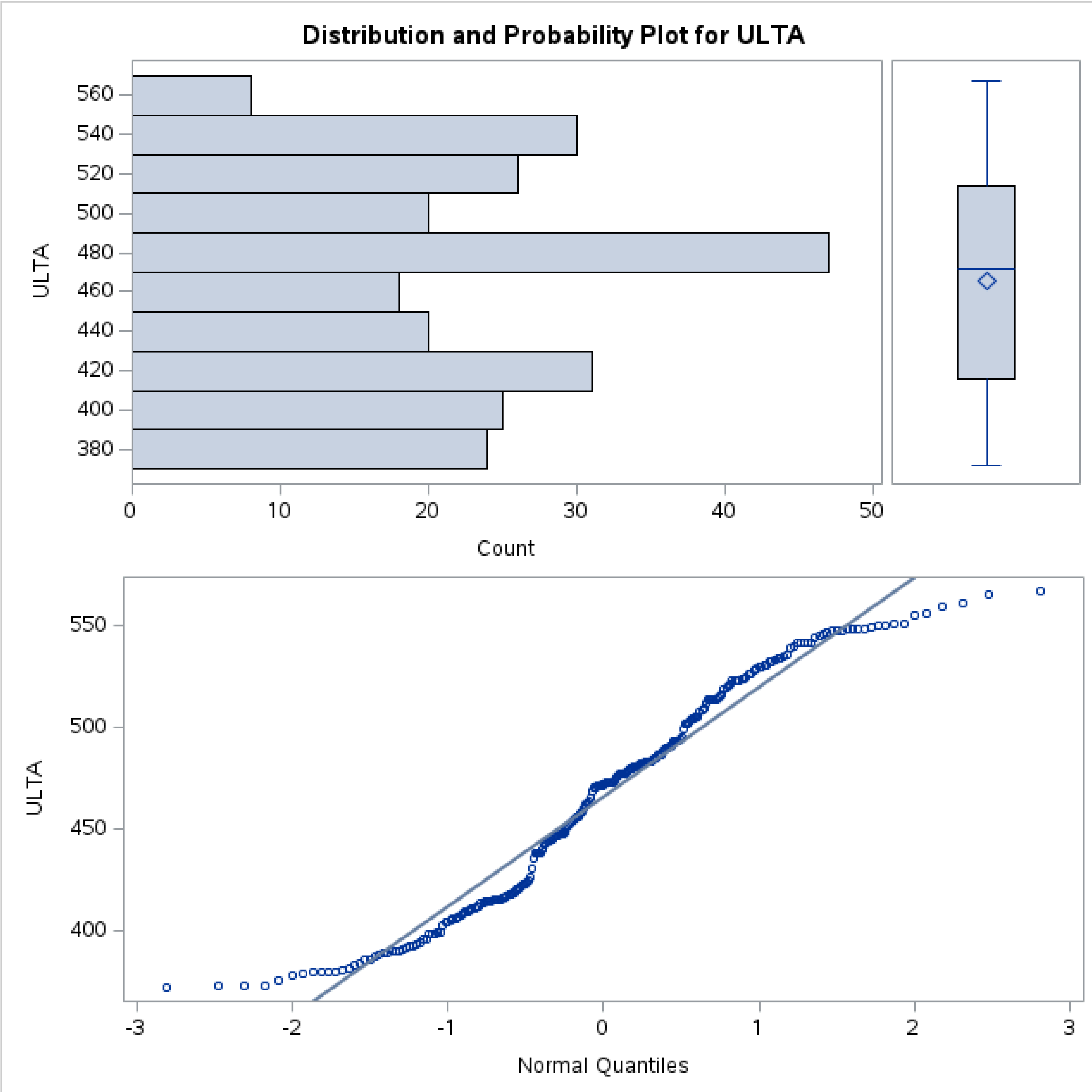
Pearson Correlation Coefficients, N = 249 Prob > |r| under H0: Rho=0		
	PHM	**RCL**
PHM **PHM**	1.00000	0.95485 <.0001
RCL **RCL**	0.95485 <.0001	1.00000

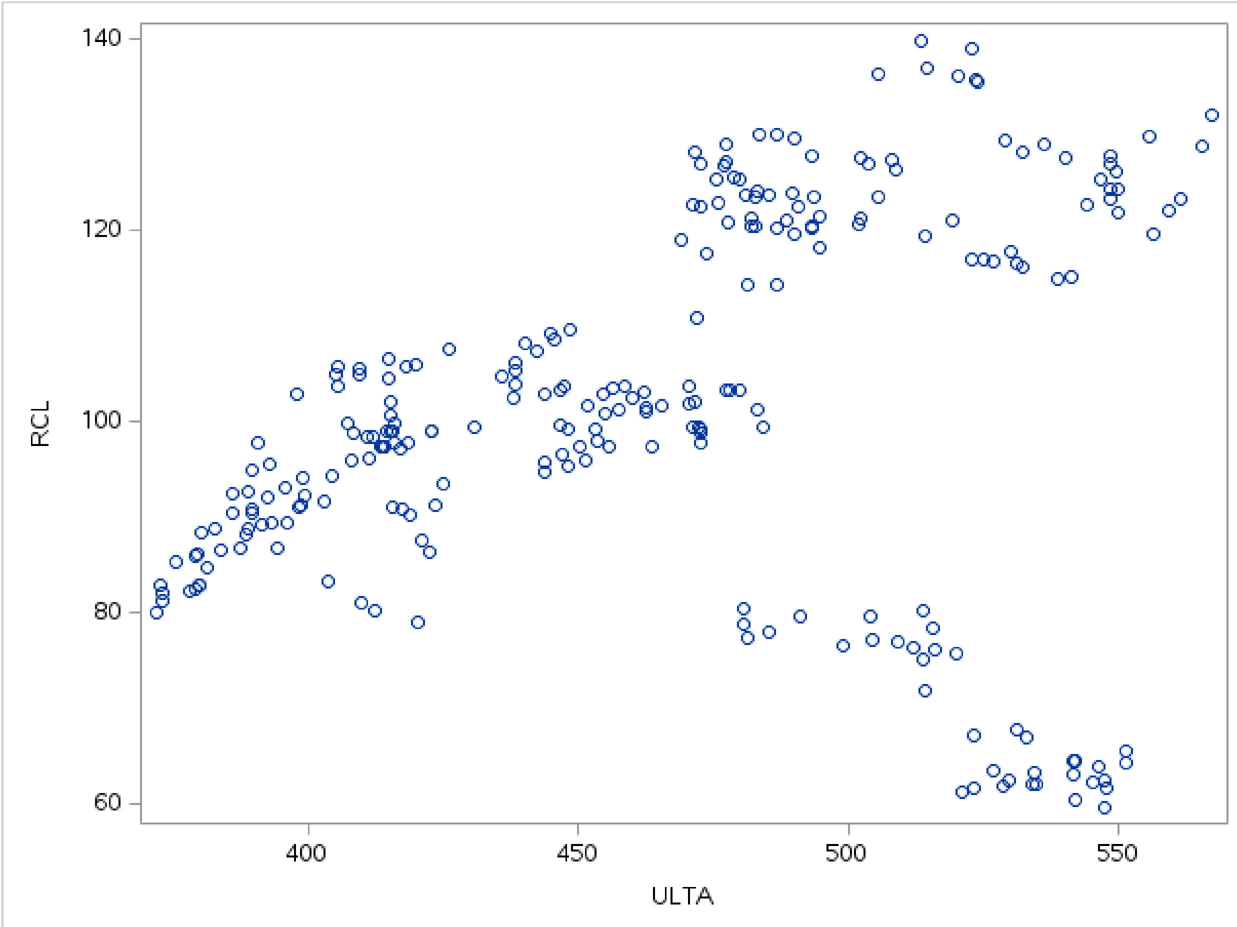

Distribution and Probability Plot for TSLA
TSLA
292.5
277.5
262.5
247.5
232.5
217.5
202.5
187.5
172.5
157.5
0
10
20
30
40
Count
300
275
250
225
200
175
150
-3
-2
-1
0
1
2
3
Normal Quantiles



TSLAR

Pearson Correlation Coefficients, N = 249 Prob > |r| under H0: Rho=0		
	TSLA	RCL
TSLA TSLA	1.00000	0.13056 0.0395
RCL RCL	0.13056 0.0395	1.00000


Distribution and Probability Plot for ULTA
ULTA
560
540
520
500
480
460
440
420
400
380
0
10
20
30
40
50
Count
ULTA
550
500
450
400
-3
-2
-1
0
1
2
3
Normal Quantiles



ULTAR

Pearson Correlation Coefficients, N = 249 Prob > |r| under H0: Rho=0		
	ULTA	RCL
ULTA ULTA	1.00000	0.20700 0.0010
RCL RCL	0.20700 0.0010	1.00000

```
proc corr data=prices;
var rcl bby bwa ccl ebay expe grmn has kmx low nclh nke phm tsla ulta;
run;
```

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	RCL	BBY	BWA	CCL
RCL RCL	1.00000	0.36345 <.0001	-0.61925 <.0001	0.83042 <.0001
BBY BBY	0.36345 <.0001	1.00000	0.29659 <.0001	0.59693 <.0001
BWA BWA	-0.61925 <.0001	0.29659 <.0001	1.00000	-0.14982 0.0180
CCL CCL	0.83042 <.0001	0.59693 <.0001	-0.14982 0.0180	1.00000
EBAY EBAY	0.23736 0.0002	0.63625 <.0001	0.11881 0.0612	0.21052 0.0008
EXPE EXPE	0.90311 <.0001	0.23697 0.0002	-0.67193 <.0001	0.70403 <.0001

48

```
proc glm data=prices_2;
model rcl = ccl | expe/solution;
store glmmodel;
proc plm restore=glmmodel noinfo;
effectplot slicefit (x=ccl sliceby=expe);
run;
```

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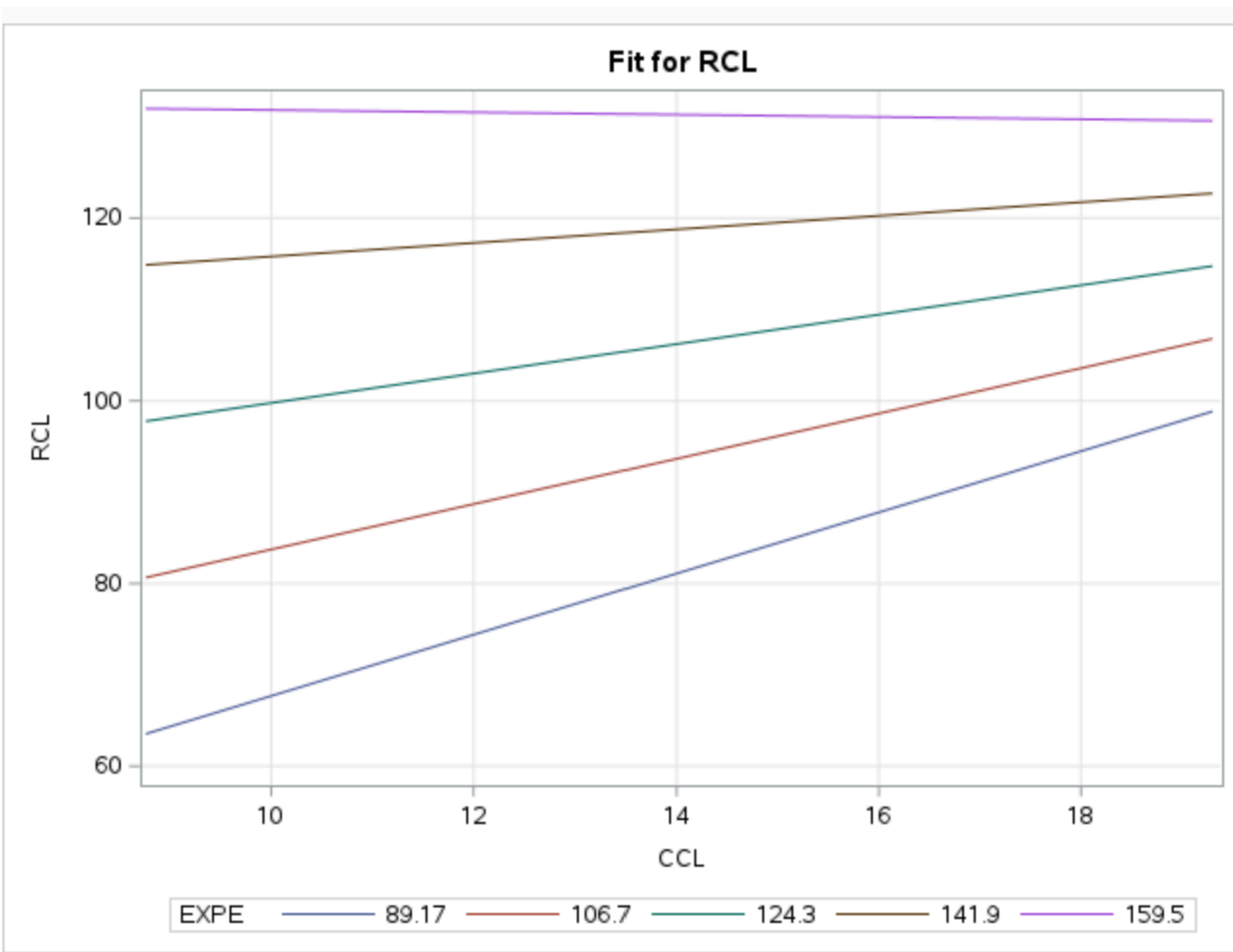
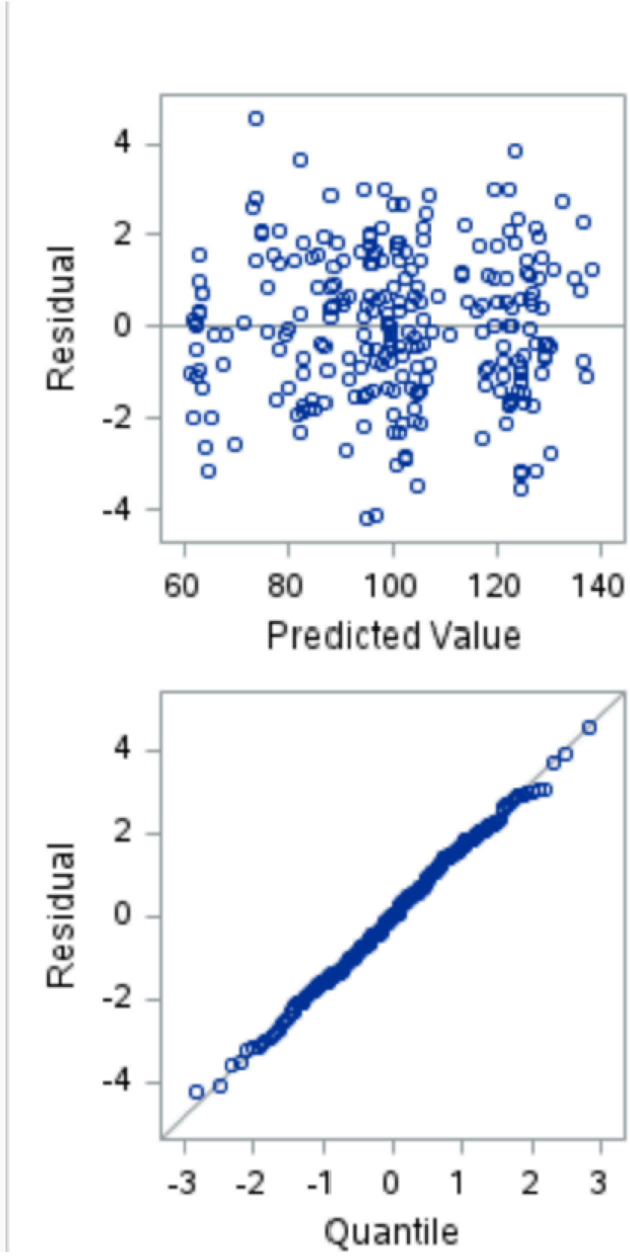


50

```
proc reg data=prices_2;
model rcl = bby bwa ccl ebay expe grmn has kmx low nclh nke phm tsla ulta;
run;
```

51



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```
proc reg data=prices_2;
model rcl = bby bwa ccl ebay expe grmn has kmx low nclh nke phm tsla ulta ccl_expe bby_sq;
test ccl_expe, bby_sq;
run;
```

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The REG Procedure
Model: MODEL1

Test 1 Results for Dependent Variable RCL				
Source	**DF**	**Mean Square**	**F Value**	**Pr > F**
Numerator	2	1.28865	0.47	0.6274
Denominator	232	2.75862		

54

```
proc reg data=prices;
model rcl = bby bwa ccl ebay expe grmn has kmx low nclh nke phm tsla ulta/vif;
run;

**phm has a crazy high vif**;
proc reg data=prices;
model rcl = bby bwa ccl ebay expe grmn has kmx low nclh nke tsla ulta/vif;
run;
**grmn looks cray**;
proc reg data=prices;
model rcl = bby bwa ccl ebay expe has kmx low nclh nke tsla ulta/vif;
run;
**take out ccl**;
proc reg data=prices;
model rcl = bby bwa ebay expe has kmx low nclh nke tsla ulta/vif;
run;

**take out HAS, insignificant p-val**;
proc reg data=prices_2;
model rcl = bby bwa ebay expe kmx low nclh nke tsla ulta/partial;
run;
```

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Parameter Estimates							
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|	Variance Inflation
Intercept	Intercept	1	72.94115	6.57995	11.09	<.0001	0
BBY	BBY	1	0.31356	0.10039	3.12	0.0020	7.49223
BWA	BWA	1	-1.57924	0.09485	-16.65	<.0001	5.97765
EBAY	EBAY	1	-0.57643	0.12039	-4.79	<.0001	4.10608
EXPE	EXPE	1	0.44319	0.02194	20.20	<.0001	6.82319
HAS	HAS	1	0.09182	0.05824	1.58	0.1162	6.49741
KMX	KMX	1	0.24021	0.07690	3.12	0.0020	10.32822
LOW	LOW	1	0.16510	0.03765	4.39	<.0001	11.37749
NCLH	NCLH	1	2.64471	0.19006	13.92	<.0001	8.40877
NKE	NKE	1	-0.20936	0.03472	-6.03	<.0001	3.22019
TSLA	TSLA	1	-0.06695	0.01114	-6.01	<.0001	5.28626
ULTA	ULTA	1	-0.06111	0.00819	-7.47	<.0001	6.58294

56

```
proc reg data=prices_2;
model rcl = bby bwa ebay expe has kmx low nclh nke tsla ulta/partial;
run;
```

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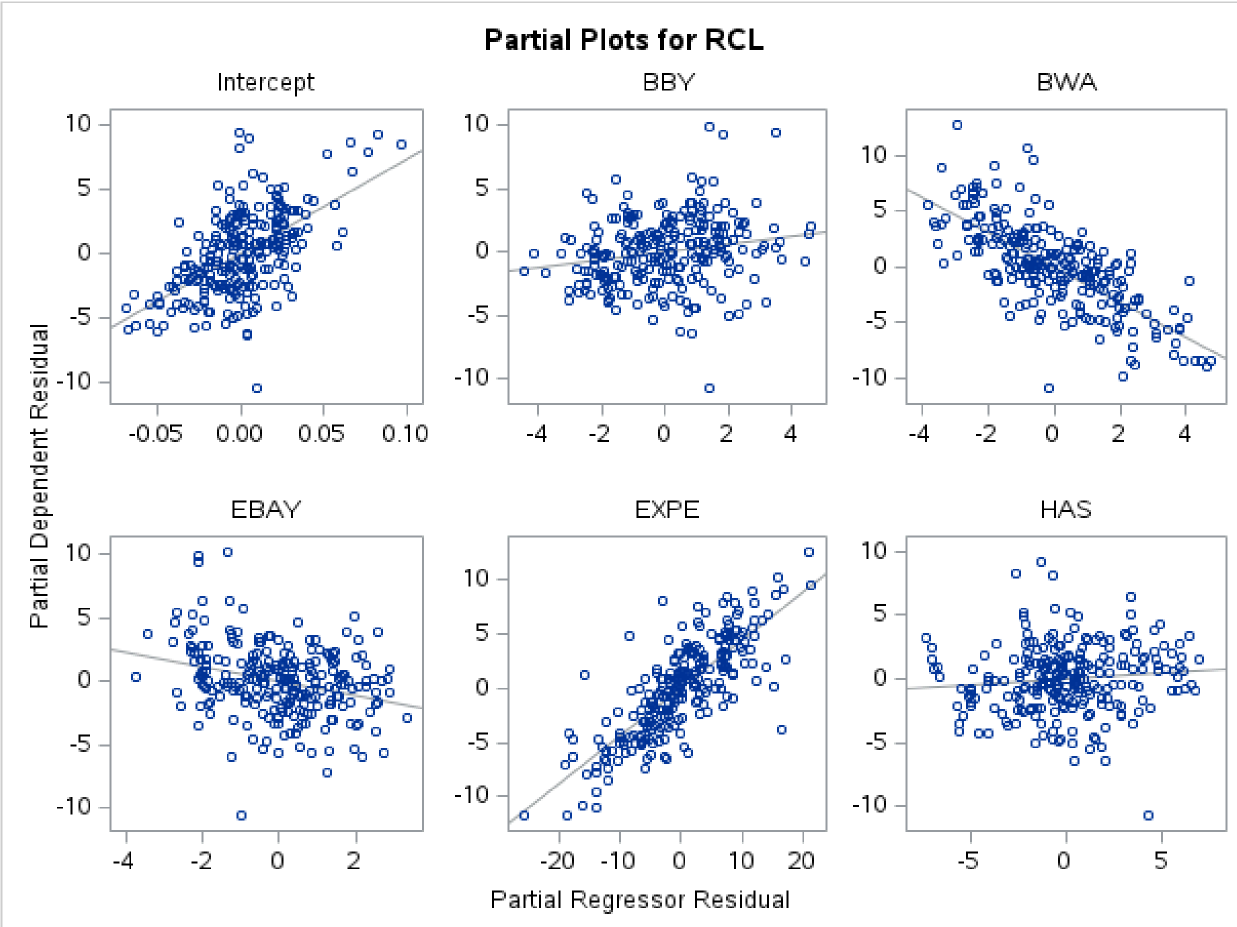
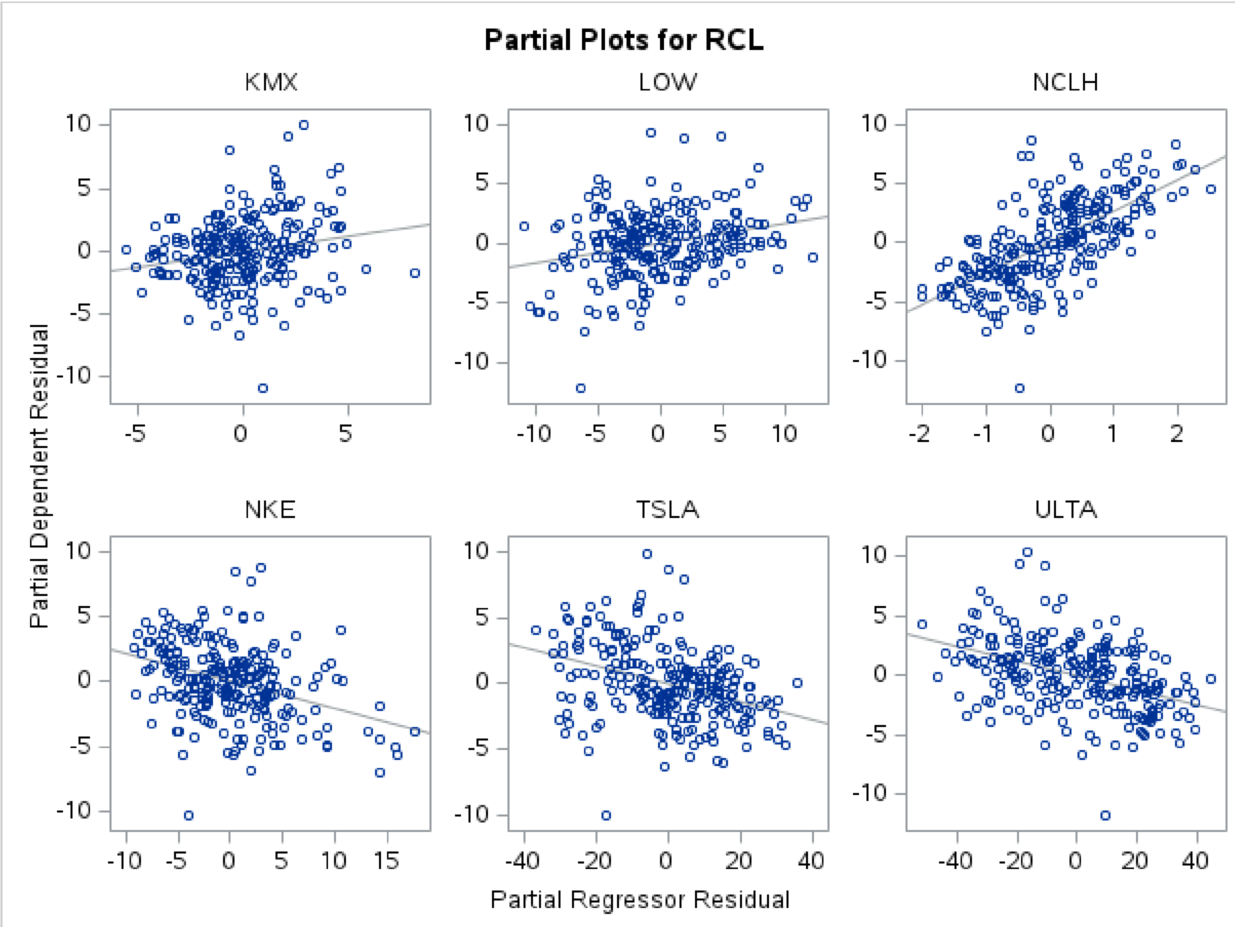

Partial Plots for RCL
Intercept
BBY
BWA
EBAY
EXPE
HAS
Partial Dependent Residual
Partial Regressor Residual



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```
proc reg data=prices;
model rcl = bby bwa ebay expe has kmx low nclh nke tsla ulta/all;
run;
```

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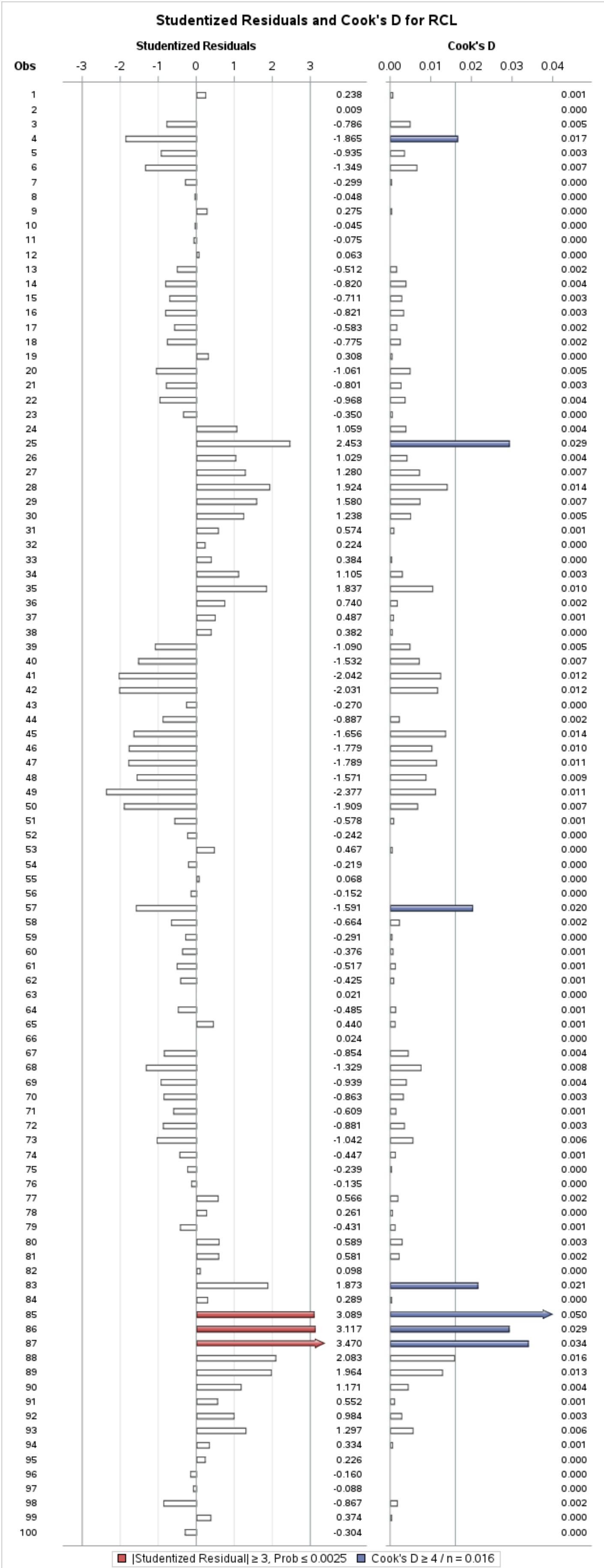
Studentized Residuals and Cook's D for RCL
Studentized Residuals
Cook's D
Obs
-3 -2 -1 0 1 2 3
0.00 0.01 0.02 0.03 0.04
1 0.238 0.001
2 0.009 0.000
3 -0.786 0.005
4 -1.865 0.017
5 -0.935 0.003
6 -1.349 0.007
7 -0.299 0.000
8 -0.048 0.000
9 0.275 0.000
10 -0.045 0.000
11 -0.075 0.000
12 0.063 0.000
13 -0.512 0.002
14 -0.820 0.004
15 -0.711 0.003
16 -0.821 0.003
17 -0.583 0.002
18 -0.775 0.002
19 0.308 0.000
20 -1.061 0.005
21 -0.801 0.003
22 -0.968 0.004
23 -0.350 0.000
24 1.059 0.004
25 2.453 0.029
26 1.029 0.004
27 1.280 0.007
28 1.924 0.014
29 1.580 0.007
30 1.238 0.005
31 0.574 0.001
32 0.224 0.000
33 0.384 0.000
34 1.105 0.003
35 1.837 0.010
36 0.740 0.002
37 0.487 0.001
38 0.382 0.000
39 -1.090 0.005
40 -1.532 0.007
41 -2.042 0.012
42 -2.031 0.012
43 -0.270 0.000
44 -0.887 0.002
45 -1.656 0.014
46 -1.779 0.010
47 -1.789 0.011
48 -1.571 0.009
49 -2.377 0.011
50 -1.909 0.007
51 -0.578 0.001
52 -0.242 0.000
53 0.467 0.000
54 -0.219 0.000
55 0.068 0.000
56 -0.152 0.000
57 -1.591 0.020
58 -0.664 0.002
59 -0.291 0.000
60 -0.376 0.001
61 -0.517 0.001
62 -0.425 0.001
63 0.021 0.000
64 -0.485 0.001
65 0.440 0.001
66 0.024 0.000
67 -0.854 0.004
68 -1.329 0.008
69 -0.939 0.004
70 -0.863 0.003
71 -0.609 0.001
72 -0.881 0.003
73 -1.042 0.006
74 -0.447 0.001
75 -0.239 0.000
76 -0.135 0.000
77 0.566 0.002
78 0.261 0.000
79 -0.431 0.001
80 0.589 0.003
81 0.581 0.002
82 0.098 0.000
83 1.873 0.021
84 0.289 0.000
85 3.089 0.050
86 3.117 0.029
87 3.470 0.034
88 2.083 0.016
89 1.964 0.013
90 1.171 0.004
91 0.552 0.001
92 0.984 0.003
93 1.297 0.006
94 0.334 0.001
95 0.226 0.000
96 -0.160 0.000
97 -0.088 0.000
98 -0.867 0.002
99 0.374 0.000
100 -0.304 0.000
|Studentized Residual| ≥ 3, Prob ≤ 0.0025
Cook's D ≥ 4 / n = 0.016

Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	Intercept	1	72.94115	6.57995	11.09	<.0001
BBY	BBY	1	0.31356	0.10039	3.12	0.0020
BWA	BWA	1	-1.57924	0.09485	-16.65	<.0001
EBAY	EBAY	1	-0.57643	0.12039	-4.79	<.0001
EXPE	EXPE	1	0.44319	0.02194	20.20	<.0001
HAS	HAS	1	0.09182	0.05824	1.58	0.1162
KMX	KMX	1	0.24021	0.07690	3.12	0.0020
LOW	LOW	1	0.16510	0.03765	4.39	<.0001
NCLH	NCLH	1	2.64471	0.19006	13.92	<.0001
NKE	NKE	1	-0.20936	0.03472	-6.03	<.0001
TSLA	TSLA	1	-0.06695	0.01114	-6.01	<.0001
ULTA	ULTA	1	-0.06111	0.00819	-7.47	<.0001